

ON THE PITFALLS OF VERBALIZED CONFIDENCE PRIORS FOR CALIBRATING LARGE REASONING MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

Large reasoning language models (LRMs) frequently suffer from overconfidence when verbalizing their confidence. Confidence-aware reinforcement learning (RL) offers a promising way to optimize calibration. However, it relies on-policy rollouts and is constrained by the pre-RL confidence distribution—termed the *confidence prior*. We reveal that base LRMs exhibit a confidence prior heavily concentrated on a few high values, and this narrow support persists throughout RL even as the specific values shift. Theoretically, we prove that this concentration restricts policy gradient updates for rarely sampled confidence values and inflates the lower bound on expected Brier risk. To overcome this exploration bottleneck, we propose **CalibSFT**, a plug-and-play supervised fine-tuning stage that shapes a calibrated confidence prior with broad support before confidence-aware RL. Specifically, CalibSFT derives confidence targets from multiple rollouts for each question by combining question-level success rates with response-level correctness, which provably preserves the optimal confidence under any strictly proper scoring rule. To avoid concentrated supervision, CalibSFT applies target-balanced sampling across target values. It then introduces correctness-conditional supervision, guiding confidence across all responses while supervising reasoning only on correct ones, so that incorrect responses encourage low confidence without reinforcing incorrect solutions. Across 16 mathematical and general reasoning benchmarks, incorporating CalibSFT consistently reduces calibration errors for 5 confidence-aware RL algorithms (e.g., reducing OOD ECE from 32.41% to 17.31% for DCPO) and improves confidence discrimination without sacrificing accuracy.

1 INTRODUCTION

Large Reasoning language models (LRMs) have made substantial progress on complex tasks such as mathematics and coding (Guo et al., 2025; Yang et al., 2025). However, these gains in accuracy do not imply reliable confidence: LRMs often remain overconfident in incorrect answers (Mei et al., 2025). A reliable model should therefore report *calibrated* confidence (Guo et al., 2017), which matches the probability that its answer is correct. Calibrated confidence indicates when an answer can be trusted, which enables applications such as hallucination detection (Wu et al., 2026) and adaptive test-time scaling (Yang et al., 2026b).

Beyond estimates derived from token probabilities, internal representations, or sampling consistency (Luo et al., 2026; Srey et al., 2026; Kuhn et al., 2023), a model can directly verbalize its confidence alongside its final answer. Unlike these alternatives, which rely on model internals or repeated sampling, verbalized confidence requires only a single black-box generation. However, base LRMs typically verbalize miscalibrated confidence that concentrates on a few high values regardless of accuracy (Xiong et al., 2024; Mei et al., 2025; Cacioli, 2026). To address this issue, post-training approaches calibrate verbalized confidence via either supervised fine-tuning (SFT) on constructed confidence targets (Xu et al., 2024; Jang et al., 2025) or confidence-aware reinforcement learning (RL) guided by calibration rewards (Damani et al., 2026; Ma et al., 2026).

Among these approaches, confidence-aware RL is particularly promising owing to its superior generalization beyond the training distribution (Chu et al., 2025). However, because confidence-aware RL learns exclusively from self-sampled responses, it hinges on the confidence distribution before

RL, which we term the *confidence prior*. In practice, this prior typically concentrates on a narrow set of high values across both correct and incorrect responses (Figure 1). Although confidence-aware RL explicitly penalizes miscalibration, this concentration persists throughout policy optimization and limits the exploration of diverse confidence values (Figure 2). As we theoretically analyze in Section 3, the policy gradient for each confidence value scales with its sampling probability, so rarely sampled values are barely updated. Such concentration in turn raises the lower bound on the expected Brier loss and thus limits calibration after RL. Empirically, post-RL confidence does not track accuracy across tasks: mean confidence on MinervaMath is only 9.7% lower than on Math500 even though Pass@1 is 39.1% lower (Figure 3).

Motivated by the miscalibration, we propose **CalibSFT**, a supervised fine-tuning stage that shapes a calibrated confidence prior with broad support before confidence-aware RL. Specifically, for each training question, CalibSFT samples multiple responses from the base model and assigns each a confidence target that balances the question-level success rate with response-level correctness. We further prove that this mixed target preserves the optimal confidence attained under correctness labels for any strictly proper scoring rule. To avoid concentrated supervision, CalibSFT employs target-balanced sampling, drawing a uniform number of responses across all target confidence values. During fine-tuning, it supervises confidence on all responses while restricting reasoning and answers to correct ones, so that incorrect responses teach the model to express low confidence without learning incorrect solutions. Since CalibSFT only reshapes the confidence prior without altering the RL objectives, it can be seamlessly applied to any confidence-aware RL framework.

We extensively evaluate CalibSFT across 16 benchmarks, spanning both in-distribution mathematical reasoning and out-of-distribution general reasoning across diverse difficulty levels. Across five confidence-aware RL methods, initializing with CalibSFT consistently outperforms base-model initialization in both calibration and discrimination without compromising task accuracy (see Table 1). For example, it reduces the out-of-distribution ECE of DCPO from 32.41% to 17.31% and improves its AUROC from 68.13% to 72.75%. Consistent with our analysis, CalibSFT mitigates confidence concentration after RL (Figure 4) and strengthens the correlation between task-level confidence and accuracy (Table 2). Furthermore, CalibSFT remains effective across different confidence formats and model families (see Tables 3 and 4).

Our contributions are summarized as follows:

1. We systematically identify that LRMs exhibit a concentrated confidence prior that persists throughout confidence-aware RL training. Theoretically, we prove that such a concentration bounds policy gradient exploration and inflates the lower bound on expected Brier loss.
2. We propose CalibSFT, a supervised fine-tuning stage that constructs confidence targets from the model’s own responses to provide a calibrated confidence prior for confidence-aware RL.
3. Extensive experiments across 16 benchmarks and 5 confidence-aware RL methods demonstrate that CalibSFT alleviates confidence concentration, boosting calibration and discrimination without compromising task accuracy.

2 PRELIMINARIES

2.1 VERBALIZED CONFIDENCE IN LARGE REASONING MODELS

In this work, we study the calibration of verbalized numerical confidence in large reasoning language models (LRMs) (Xiong et al., 2024; Damani et al., 2026). Given a question x , the LRM π_θ generates a response $o = (r, a, c)$, where r is a reasoning trace, a is an answer, and $c \in [0, 1]$ is a post-answer confidence value that represents the model’s estimated probability that a is correct.

Confidence-aware rewards. Reinforcement learning with verifiable rewards (RLVR) trains the policy with the binary correctness label $z = V(x, a) \in \{0, 1\}$ of an automated verifier V as the reward. To further incorporate confidence into the optimization objective, confidence-aware RL adds a calibration reward that depends on both c and z . For example, RLCR (Damani et al., 2026) sets the calibration reward to the negative Brier loss. The resulting total reward favors correct answers and, for a given correctness, prefers confidence closer to z :

$$R(o) = z - (c - z)^2. \quad (1)$$

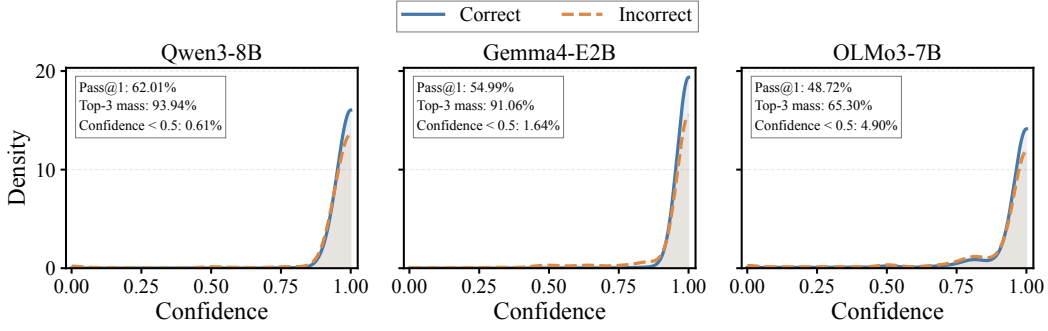


Figure 1: Confidence distributions from three base models on DeepScaleR-Train. Confidence concentrates on a few high values for both correct and incorrect responses.

Policy optimization. Confidence-aware RL typically optimizes this reward with Group Relative Policy Optimization (GRPO) (Shao et al., 2024). For each question, GRPO samples G responses $\{o_i\}_{i=1}^G$ from the old policy $\pi_{\theta_{\text{old}}}$ and normalizes their rewards $R_i = R(o_i)$ within the group to obtain the advantage

$$\hat{A}_i = \frac{R_i - \bar{R}}{\sigma_R + \epsilon}, \quad \bar{R} = \frac{1}{G} \sum_{j=1}^G R_j, \quad \sigma_R = \text{std}(\{R_j\}_{j=1}^G), \quad (2)$$

where $\epsilon > 0$ ensures numerical stability. GRPO then updates the policy by maximizing

$$J_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{L_i} \sum_{t=1}^{L_i} \min \left(\rho_{i,t}(\theta) \hat{A}_i, \text{clip}(\rho_{i,t}(\theta), 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}}) \hat{A}_i \right) \right], \quad (3)$$

where $o_{i,t}$ is the t -th token of o_i , L_i is the length of o_i , $\rho_{i,t}(\theta) = \pi_{\theta}(o_{i,t} \mid x, o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} \mid x, o_{i,<t})$ is the importance ratio, and ϵ_{clip} is the clipping parameter.

2.2 CALIBRATION METRICS

A useful verbalized confidence score should be both calibrated and discriminative (Eusebi et al., 2026). Let C and Z denote the confidence and correctness random variables. Confidence is perfectly calibrated if $\Pr(Z = 1 \mid C = p) = p$ for every $p \in [0, 1]$, and it is discriminative if correct responses receive higher confidence than incorrect ones. We typically use **Expected Calibration Error (ECE)** (Guo et al., 2017) to measure the deviation from perfect calibration $\mathbb{E}_C[|\Pr(Z = 1 \mid C) - C|]$. After partitioning N responses into B confidence bins $\{\mathcal{I}_b\}_{b=1}^B$, we estimate it as:

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{I}_b|}{N} |\text{acc}(\mathcal{I}_b) - \text{conf}(\mathcal{I}_b)|, \quad (4)$$

where $\text{acc}(\mathcal{I}_b)$ and $\text{conf}(\mathcal{I}_b)$ are the mean correctness and confidence in bin b . While ECE assesses calibration at the bin level, the **Brier score** (Brier, 1950) computes response-level squared error, $\frac{1}{N} \sum_{j=1}^N (c_j - z_j)^2$, combining calibration with resolution (i.e., how well confidence separates correct and incorrect responses) (Murphy, 1973). However, calibration alone does not imply discrimination. For example, assigning every response a constant confidence equal to the overall accuracy yields zero ECE while giving correct and incorrect responses identical scores (Eusebi et al., 2026). We therefore also report **AUROC** to evaluate threshold-agnostic discrimination between correct and incorrect answers. These complementary metrics jointly evaluate calibration and discrimination, and we next investigate these properties in base LRMs and throughout RL training.

3 MOTIVATION

Confidence-aware RL typically adds a calibration reward that scores each response’s confidence against its correctness (Xu et al., 2024; Damani et al., 2026; Ma et al., 2026; Yang et al., 2026b).

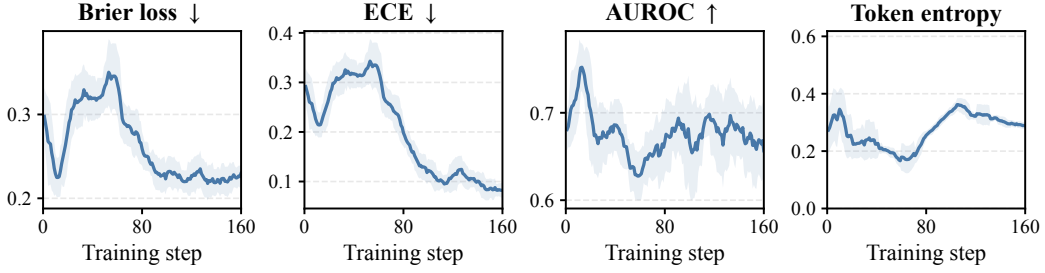


Figure 2: Training dynamics of RLCR on Qwen3-8B. The calibration reward lowers Brier loss and ECE, but discrimination (AUROC) also declines because the confidence token entropy fails to rise, restricting the policy’s exploration over confidence values.

Whether this reward alone produces well-calibrated confidence depends on the distribution of sampled confidence values. We therefore examine this distribution before RL, during training, and at test time.

Confidence priors concentrate on a few high values. We refer to the confidence distribution before RL as the model’s *confidence prior*. To characterize this prior, we sample 10 responses per question from Qwen3-8B (Yang et al., 2025), Gemma4-E2B-Instruct (Abd et al., 2026), and OLMo3-7B-Instruct (Ettinger et al., 2025) on DeepScaleR-Train. Figure 1 presents their confidence distributions for correct and incorrect responses. All three models place most of their confidence mass on a few high values, with fewer than 5% of outputs below 0.5. Confidence remains high even for incorrect responses, so it does not track answer correctness. RL therefore starts from a concentrated confidence prior that rarely produces low-confidence samples.

RL improves calibration but confidence concentration persists. To examine whether confidence-aware RL overcomes this concentration, we train Qwen3-8B with RLCR (Damani et al., 2026). As shown in Figure 2, although the calibration reward lowers Brier loss and ECE, discrimination (AUROC) also declines. This decline is accompanied by persistently low confidence token entropy, which prevents the policy from expanding its exploration over diverse confidence values. Consequently, the model merely shifts probability mass to satisfy average calibration, without learning to distinguish correct and incorrect responses. We provide results across more confidence-aware RL algorithms in Appendix D.2, where confidence token entropy consistently tracks the number of distinct sampled confidence values throughout training.

To better understand why this concentration persists during training, we formulate confidence generation as action selection under a softmax policy (Garg et al., 2022). Since the LLM decodes each confidence value autoregressively, we model the induced distribution over the finite set of confidence values as a softmax over per-value logits.

Proposition 1 (Policy gradients under confidence concentration). *Given a question x and a finite set of confidence actions $\{c_1, \dots, c_K\} \subseteq [0, 1]$, let $\eta = (\eta_1, \dots, \eta_K)$ denote the policy’s logits for these actions and $\pi_i = \Pr(C = c_i \mid x) = \exp(\eta_i) / \sum_{j=1}^K \exp(\eta_j)$. Define the fixed expected reward $u_i = -\mathbb{E}[(c_i - Z)^2 \mid x]$. Let $J_x(\eta) = \sum_i \pi_i u_i$ and $\Delta_x = \max_i u_i - \min_i u_i$. Then*

$$\frac{\partial J_x}{\partial \eta_i} = \pi_i(u_i - J_x), \quad \left| \frac{\partial J_x}{\partial \eta_i} \right| \leq \pi_i \Delta_x. \quad (5)$$

If $\pi_k = 1 - \varepsilon$ for some $k \in \{1, \dots, K\}$, then

$$\|\nabla_{\eta} J_x\|_1 \leq 2\varepsilon \Delta_x. \quad (6)$$

Proposition 1 shows that the logit gradient for any confidence action c_i is directly scaled by its sampling probability π_i (Eq. 5). Consequently, a rarely sampled confidence value receives a negligible gradient even when choosing it would yield a higher expected reward. Furthermore, when the policy heavily concentrates on a single value ($\pi_k = 1 - \varepsilon$ with $\varepsilon \ll 1$), Eq. 6 bounds the overall logit gradient norm by $2\varepsilon \Delta_x$. As $\varepsilon \rightarrow 0$, the entire logit gradient vanishes, preventing the model from escaping a suboptimal confidence prior. A concentrated confidence distribution therefore impedes

exploration of better-calibrated outputs, consistent with the persistently low confidence token entropy observed in Figure 2. We provide the proof in Appendix B.1.

Persistent confidence concentration hinders downstream calibration.

We evaluate the final RLCR checkpoint on Math500 and MinervaMath, two datasets of contrasting difficulty. As shown in Figure 3, the post-RL model underestimates its accuracy on Math500 (90.8% Pass@1 vs. 84.5% mean confidence) but is overconfident on MinervaMath (51.7% Pass@1 vs. 74.8% mean confidence). Despite a 39.1-point gap in Pass@1 between the two tasks, mean confidence differs by only 9.7 points. The three most frequent values still account for 82.25% and 69.49% of outputs on Math500 and MinervaMath, respectively. The same small set of confidence values therefore persists across tasks of different difficulty, so confidence no longer tracks the model’s accuracy on each task.

To characterize the cost of this miscalibration, we bound the expected Brier loss $\mathbb{E}[(C - Z)^2 | x]$ (Gneiting & Raftery, 2007), which is minimized at $C = p_x$, where $p_x = \Pr(Z = 1 | x)$.

Proposition 2 (Brier risk under limited confidence support). *Assume that C and Z are conditionally independent given x , consistent with the limited discrimination between correct and incorrect responses observed in Figure 1. For any $\delta > 0$, let $m_\delta(x) = \Pr(|C - p_x| < \delta | x)$. Then*

$$\mathbb{E}[(C - Z)^2 | x] \geq p_x(1 - p_x) + \delta^2(1 - m_\delta(x)). \quad (7)$$

Proposition 2 decomposes the expected Brier loss into the irreducible uncertainty $p_x(1 - p_x)$, which no confidence policy can reduce, and an excess penalty $\delta^2(1 - m_\delta(x))$ (Eq. 7). When the confidence distribution concentrates far from the true success probability p_x , the probability mass around p_x vanishes ($m_\delta(x) \rightarrow 0$) and this penalty rises to δ^2 , so retaining only values distant from p_x inflates the best achievable Brier loss no matter how the calibration reward reweights them. In Appendix B.2, we provide the detailed proof, extend the bound to a restricted confidence support, and show that limited support similarly raises the expected error of the confidence value closest to p_x among the G responses sampled per question.

4 METHOD

Section 3 showed that confidence concentration persists during RL, since rarely sampled confidence values have small logit gradients, and that the resulting narrow support hinders downstream calibration. Our key idea is therefore to set the confidence prior by supervision before RL, rather than leaving it to exploration. We propose CalibSFT, a supervised fine-tuning stage for confidence initialization. For each question, CalibSFT samples multiple responses from the base model and assigns each a confidence target that combines the question’s success rate with the response’s own correctness. It supervises confidence on all responses, but reasoning and answers only on correct ones.

4.1 PROPER-SCORING CONFIDENCE TARGETS

Under a proper scoring rule, the optimal confidence equals the probability that the answer is correct (Gneiting & Raftery, 2007). Since this probability is not observed directly, we estimate it from the base model’s own responses. For each training question x , we sample n responses with correctness labels $z_i \in \{0, 1\}$ and use their success rate as the estimate. However, correct and incorrect responses to the question then share the same confidence target. To support discrimination, we combine the success rate with the response-level label z_i and construct the confidence target of response i as

$$q_x = \frac{1}{n} \sum_{i=1}^n z_i, \quad c_i^* = \lambda q_x + (1 - \lambda) z_i, \quad (8)$$

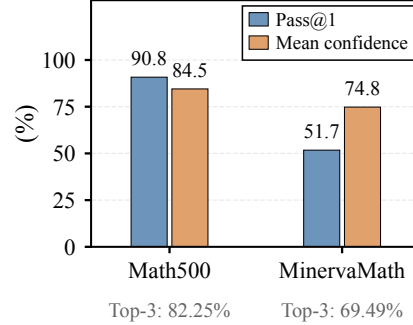


Figure 3: Performance comparison on Math500 and MinervaMath of RLCR on Qwen3-8B. Mean confidence shifts far less than Pass@1 between the two tasks.

where the mixing weight $\lambda \in [0, 1]$ balances the two signals. The mixed target is no longer a correctness label, so we examine whether it still yields the same optimum under a strictly proper scoring rule. Let $p_x = \Pr(Z = 1 \mid x)$ under the base model, and let Z_i and C_i^* denote the random counterparts of z_i and c_i^* when the n responses are sampled independently. We then have the following proposition.

Proposition 3 (Proper-scoring optimum of the mixed target). *Let $\ell(a, z)$ be a strictly proper scoring loss for $z \in \{0, 1\}$, extended to $t \in [0, 1]$ by $\ell(a, t) = t\ell(a, 1) + (1-t)\ell(a, 0)$. Then $\mathbb{E}[C_i^* \mid x] = p_x$, and for every a at which ℓ is finite,*

$$\mathbb{E}[\ell(a, C_i^*) \mid x] = \mathbb{E}[\ell(a, Z_i) \mid x]. \quad (9)$$

Proposition 3 shows that the mixed target is equivalent to the correctness label in expected loss, so p_x remains the optimal prediction under any strictly proper ℓ . The number of responses n controls the variance of the success-rate estimate and the weight λ controls the target’s discrimination within a question; we set $n = 50$ and $\lambda = 0.5$ by default. We prove the proposition in detail and derive the variance of C_i^* in Appendix B.3.

In practice, we replace each response’s confidence with c_i^* , preserving its reasoning and answer. Since the target values are unevenly distributed across questions, training on the raw data would again concentrate supervision on a few target values. We therefore apply target-balanced sampling, which draws the same number of responses for each target value (Appendix C.1).

4.2 CORRECTNESS-CONDITIONAL SUPERVISION

The constructed data contain both correct and incorrect responses. The incorrect ones are kept because their lower targets teach the model to express low confidence in its own errors, whereas imitating their reasoning would train the model on incorrect solutions. CalibSFT therefore supervises confidence on all responses and restricts reasoning and answer supervision to correct ones.

Formally, let $\tilde{o}$ denote a response whose confidence has been replaced by its constructed target, and let $\mathcal{T}_{\text{ans}}$ and $\mathcal{T}_{\text{conf}}$ denote the positions of its reasoning and answer tokens and its confidence tokens, respectively. The CalibSFT objective is then

$$\begin{aligned} \mathcal{L}_{\text{CalibSFT}} = & (1 - \beta) \mathbb{E}_{\text{correct}} \left[-\frac{1}{|\mathcal{T}_{\text{ans}}|} \sum_{t \in \mathcal{T}_{\text{ans}}} \log \pi_{\theta}(\tilde{o}_t \mid x, \tilde{o}_{<t}) \right] \\ & + \beta \mathbb{E}_{\text{all}} \left[-\frac{1}{|\mathcal{T}_{\text{conf}}|} \sum_{t \in \mathcal{T}_{\text{conf}}} \log \pi_{\theta}(\tilde{o}_t \mid x, \tilde{o}_{<t}) \right]. \end{aligned} \quad (10)$$

Here $\mathbb{E}_{\text{correct}}$ averages over the correct responses and $\mathbb{E}_{\text{all}}$ over all of them. Averaging within each segment prevents the longer reasoning sequences from dominating the loss by token count. The weight β balances confidence supervision against reasoning supervision, and we set $\beta = 0.5$ so that the two terms contribute equally.

Integration with confidence-aware RL. CalibSFT only changes the confidence prior and leaves the RL objective unchanged, so it can be applied before any confidence-aware RL method. We evaluate it across RL methods in Section 5.2 and analyze its effect on confidence concentration after RL in Section 5.3. We further examine its generality across confidence formats and base models in Section 5.4. Finally, we ablate the confidence target construction, the mixing weight λ , and the correctness-conditional supervision in Appendix D.4.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines. We compare methods spanning answer-only and confidence-aware training. **Base** denotes the checkpoint without task-specific training, while **RLVR** optimizes only answer correctness. We use confidence-aware RL baselines including **RLCR** (Damani et al., 2026), **ReDoubt** (Bani-Harouni et al., 2026), **CoCA** (Li et al., 2026a), **DCPO** (Ma et al., 2026), and **C3RL** (Yang et al.,

2026b). To measure the effect of **CalibSFT**, we evaluate each confidence-aware RL method from both the Base and CalibSFT checkpoints. All methods use Qwen3-8B (Yang et al., 2025) as the default model and follow the **DAPO** training recipe (Yu et al., 2025).

Datasets. We use **DeepScaleR-Preview** (Luo et al., 2025) as the mathematical training dataset. We evaluate on a broad range of in-distribution (ID) and out-of-domain (OOD) datasets spanning different levels of difficulty. For ID mathematical reasoning, we use **DeepScaleR-Eval**, **MATH-500** (Hendrycks et al., 2021), **MinervaMath** (Lewkowycz et al., 2022), **OlympiadBench** (He et al., 2024), **GSM8K** (Cobbe et al., 2021), and **AIME 2024–2026** (Dekoninck et al., 2026). For general reasoning, we use **HotpotQA** (Yang et al., 2018), **MuSiQue** (Trivedi et al., 2022), **TriviaQA** (Joshi et al., 2017), **NQOpen** (Lee et al., 2019), **PopQA** (Mallen et al., 2023), **WebQuestions** (Berant et al., 2013), **DROP** (Dua et al., 2019), and **LiveBenchReasoning** (White et al., 2024).

Evaluation metrics. We evaluate model performance using complementary metrics that capture answer correctness, confidence discrimination, and absolute calibration. We report **Pass@1** for answer correctness, **AUROC** for ranking correct responses above incorrect responses, and **Brier score** (Brier, 1950) and **ECE** (Guo et al., 2017) for probabilistic calibration.

Implementation details. During evaluation, we use temperature 0.6 and sample 32 responses per problem for the AIME benchmarks and 4 responses per problem for all other benchmarks. All experiments are conducted on a single node with eight NVIDIA B20Z GPUs. We report our hyperparameter setting and training details in Appendix C.

5.2 OVERALL RESULTS

CalibSFT consistently benefits diverse confidence-aware RL methods. As shown in Table 1, initializing RL from CalibSFT reduces Brier score and ECE and increases AUROC across all five confidence-aware RL baselines on both mathematical and general-reasoning benchmarks. As a concrete example, CalibSFT reduces DCPO’s ECE from 29.05 to 8.83 while lifting its AUROC from 63.57 to 76.00 on MinervaMath. Moreover, these calibration gains come without sacrificing accuracy. The average Pass@1 improves by 1.20 points in-distribution and 1.49 points out-of-distribution across the five methods. We provide detailed per-dataset results in Table 6-7 (Appendix D.1). Overall, these results establish CalibSFT as a plug-and-play initialization that consistently strengthens confidence-aware RL for better calibration.

CalibSFT and RL contribute complementary gains. To disentangle their contributions, we compare CalibSFT and RLVR individually. CalibSFT alone substantially mitigates overconfidence, with ECE dropping from 34.36 to 14.23 in-distribution and from 46.94 to 25.49 out-of-distribution. In contrast, Base and RLVR remain severely miscalibrated, indicating that answer correctness alone is insufficient to reshape confidence. However, calibration does not imply better confidence ranking: CalibSFT’s OOD AUROC even declines from 65.71 to 61.20, reflecting the limited generalization of supervised training. Confidence-aware RL fills exactly this gap by directly optimizing discrimination, and combining it with CalibSFT yields the strongest overall performance.

5.3 CALIBRATION ANALYSIS

CalibSFT aligns confidence with accuracy across tasks. A well-calibrated model should express higher confidence on tasks it can reliably solve and lower confidence on harder ones. To test this, we compute Pearson and Spearman correlations between per-dataset mean confidence and per-dataset accuracy for each RL method on the ID and OOD splits separately. As shown in Table 2, CalibSFT improves Spearman correlation for all RL baselines on both mathematical and general reasoning benchmarks. For example, DCPO’s Spearman correlation rises from 0.619 to 0.976 on mathematical tasks and from 0.548 to 0.643 on general reasoning. Pearson correlation also improves consistently across all baselines and splits (e.g., from 0.505 to 0.729 for RLVR on OOD). These results indicate that CalibSFT helps the model assign higher confidence to tasks it is more likely to answer correctly.

Table 1: Performance comparison on in-distribution and out-of-distribution benchmarks. Each entry averages the datasets within the corresponding split. CalibSFT initialization achieves notably better calibration, discrimination, and Pass@1 for all RL baseline.

Method	In-distribution				Out-of-distribution			
	Pass@1 $\uparrow$	AUROC $\uparrow$	Brier $\downarrow$	ECE $\downarrow$	Pass@1 $\uparrow$	AUROC $\uparrow$	Brier $\downarrow$	ECE $\downarrow$
Base	59.32	71.90	33.20	34.36	45.36	65.71	45.15	46.94
RLVR	64.16	73.16	31.10	31.52	43.88	64.19	50.29	51.52
CalibSFT	58.51	79.93	17.30	14.23	46.37	61.20	29.15	25.49
RLCR	65.33	82.36	14.43	14.46	43.96	69.38	31.14	31.17
+ CalibSFT	67.20	86.32	12.77	8.59	45.71	71.31	23.24	18.20
ReDoubt	65.59	81.05	15.48	16.94	43.71	69.88	25.07	23.98
+ CalibSFT	67.59	86.99	12.90	14.28	45.55	73.93	22.42	18.89
CoCA	63.62	80.66	16.31	19.95	43.68	66.79	26.67	24.12
+ CalibSFT	65.07	85.63	16.06	18.51	45.55	71.22	22.94	18.89
DCPO	64.68	78.21	14.69	12.16	43.99	68.13	32.85	32.41
+ CalibSFT	64.84	85.08	12.83	10.39	44.82	72.75	21.73	17.31
C3RL	65.62	68.36	25.44	28.06	43.44	62.19	45.54	48.00
+ CalibSFT	66.17	79.70	15.91	14.35	44.60	63.03	23.89	17.38

Table 2: Pearson r and Spearman ρ correlation between accuracy and mean confidence across ID and OOD datasets. Both correlations increase for all RL methods when incorporating CalibSFT, indicating that confidence better tracks task-level accuracy.

Dataset	Method	RLCR		ReDoubt		CoCA		DCPO		C3RL	
		r	ρ	r	ρ	r	ρ	r	ρ	r	ρ
ID	Baseline	0.768	0.619	0.829	0.762	0.750	0.643	0.669	0.619	0.635	0.595
	+CalibSFT	0.963	0.952	0.976	0.976	0.887	0.976	0.996	0.976	0.896	0.833
OOD	Baseline	0.505	0.548	0.667	0.762	0.447	0.571	0.467	0.548	0.681	0.714
	+CalibSFT	0.729	0.619	0.738	0.881	0.605	0.619	0.611	0.643	0.848	0.929

CalibSFT mitigates downstream confidence concentration after RL.

Prior work observes that confidence estimates often concentrate on a few dominant values even after confidence-aware RL training (Wang et al., 2026; Yang et al., 2026b). To examine whether CalibSFT initialization alleviates this concentration, we compare RLCR, CoCA, and DCPO on MinervaMath, with and without CalibSFT. As shown in Figure 4, all three methods trained from Base exhibit heavy peaks at a few high-confidence values, while initializing from CalibSFT flattens these peaks and aligns the mean confidence with the actual accuracy. As a result, CalibSFT alleviates both concentration and overconfidence and lowers ECE for all three methods. For example, DCPO’s dominant peak near high confidence is redistributed toward lower values while accuracy is preserved, lowering ECE from 29.05% to 15.58%. These results suggest that CalibSFT initialization not only calibrates the SFT stage but also propagates lower concentration and better calibration into the downstream RL policy.

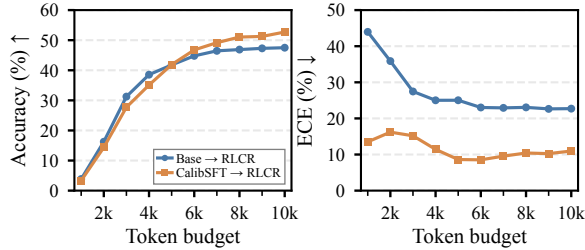


Figure 5: Accuracy ($\uparrow$) and ECE ($\downarrow$) on AIME’26 across token budgets for RLCR. CalibSFT initialization yields lower ECE across all budgets while maintaining accuracy.

The calibration gains of CalibSFT persist across reasoning budgets. A useful calibrator should keep confidence reliable regardless of how much test-time reasoning the model is allocated. To test this, we compare RLCR with CalibSFT on AIME 2026 across a range of token budgets. As shown

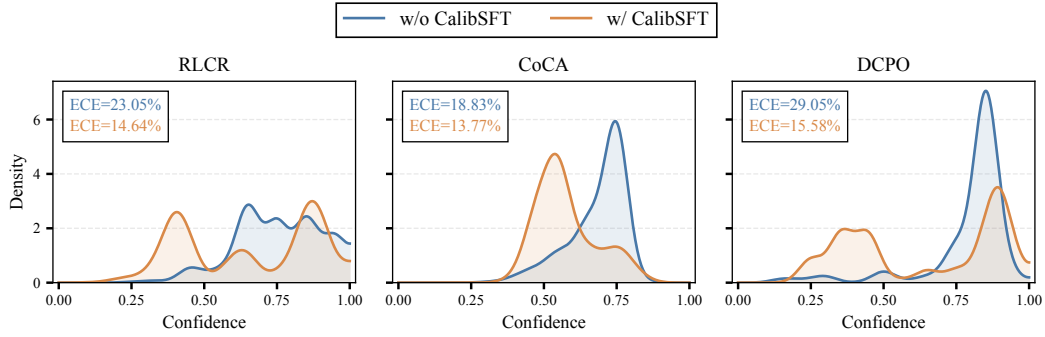


Figure 4: Confidence distributions on MinervaMath for RLCR, CoCA, and DCPO. CalibSFT reduces concentration at dominant confidence values and lowers ECE for all three methods.

Table 3: AUROC ($\uparrow$) and ECE ($\downarrow$) of Qwen3-8B across confidence formats. “Base” denotes the model without training and “+Ours” the model after CalibSFT. Values are averaged over the eight ID or eight OOD datasets.

Confidence Format	ID				OOD			
	ECE $\downarrow$		AUROC $\uparrow$		ECE $\downarrow$		AUROC $\uparrow$	
	Base	+Ours	Base	+Ours	Base	+Ours	Base	+Ours
Probability $\{0.00, 0.01, \dots, 1.00\}$	14.05	8.59	81.81	86.32	32.81	18.20	68.72	71.31
Probability $\{0.00, 0.05, \dots, 1.00\}$	14.05	12.13	81.81	86.05	32.81	19.30	68.72	71.23
Integer (0–9)	14.69	18.66	79.00	84.14	26.31	16.70	67.00	69.36
Linguistic (low / medium / high)	17.71	15.90	73.56	79.58	24.06	20.31	64.20	67.73

in Figure 5, CalibSFT achieves substantially lower ECE at every budget while accuracy remains comparable to the RLCR baseline. For example, at a 10,000-token budget, ECE drops from 22.71 to 10.97 without sacrificing accuracy. These results indicate that CalibSFT’s calibration gains persist across reasoning budgets.

5.4 GENERALITY OF CALIBSFT

CalibSFT is effective across diverse confidence formats. We further show that CalibSFT is effective across different ways of expressing confidence. Specifically we train CalibSFT with four confidence target formats: two-decimal probabilities on a fine grid $\{0.00, 0.01, \dots, 1.00\}$ or a coarse grid $\{0.00, 0.05, \dots, 1.00\}$, integer scores from 0 to 9, and linguistic labels (*low*, *medium*, *high*) (See Appendix C.3 for details). As shown in Table 3, CalibSFT consistently improves both ECE and AUROC across all four formats, indicating that its benefit does not rely on a particular way of expressing confidence. The reduction in ECE is largest with the fine probability grid, whose granularity lets confidence approximate the probability of correctness most closely.

CalibSFT is agnostic to the base model. To examine the transferability of CalibSFT across model architectures, we train Gemma4-E2B (Abd et al., 2026) using RLCR with CalibSFT, and then evaluate it on three ID and three OOD benchmarks spanning a range of difficulty. As shown in Table 4, CalibSFT improves both ECE and AUROC on all benchmarks, halving the average ECE from 23.41 to 12.49 and raising the average AUROC from 70.49 to 73.95. These results indicate that the calibration benefit of CalibSFT holds across model families on benchmarks of varying difficulty.

Table 4: AUROC ($\uparrow$) and ECE ($\downarrow$) of RLCR on Gemma4-E2B across benchmarks of varying difficulty. CalibSFT noticeably improves calibration across all difficulty levels.

Metric	Method	In-distribution			Out-of-distribution			Average
		Math500	Olympiad	Minerva	DROP	HotpotQA	NQOpen	
ECE $\downarrow$	RLCR	7.84	17.69	30.34	19.92	26.41	38.25	23.41
	+CalibSFT	5.34	7.76	12.74	13.30	13.04	22.77	12.49
AUROC $\uparrow$	RLCR	75.47	83.24	66.29	60.60	60.48	76.85	70.49
	+CalibSFT	80.34	87.35	70.75	65.45	62.66	77.13	73.95

6 CONCLUSION

[TODO:] Summarize the concentrated and misaligned confidence prior, the rollout-derived and target-balanced CalibSFT initialization, and its effect on downstream confidence-aware RL. Close with the main ID/OOD evidence and the remaining limitations of rollout-derived confidence targets.

AI USE STATEMENT

(This section is **required** and does not count toward the page limit.)

In this work, we used generative AI tools for [tasks with required disclosure]. We have not used generative AI tools for [other tasks with required disclosure], and [the rest of the required disclosure tasks] are not applicable to this work. Additionally, we used generative AI tools for [tasks with recommended disclosure]. We have reviewed all AI-assisted work. [Elaborate. For example, “we checked LLM-generated research ideas for potential plagiarism through a manual literature survey”, “LLM-generated code was verified and tested for correctness by 2 authors”, etc.]. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

See the ICLR 2027 AI Policy for Authors for more details. This statement should not be more than 1 page.

ETHICS STATEMENT

(This section is **recommended** and does not count toward the page limit.)

If authors feel that their paper submission raises questions regarding the Code of Ethics, they are encouraged to include a paragraph of Ethics Statement (at the end of the main text before references) to address potential concerns where appropriate. Topics include, but are not limited to, studies that involve human subjects, practices to data set releases, potentially harmful insights, methodologies and applications, potential conflicts of interest and sponsorship, discrimination/bias/fairness concerns, privacy and security issues, legal compliance, and research integrity issues (e.g., IRB, documentation, research ethics). This statement should not be more than 1 page.

REPRODUCIBILITY STATEMENT

(This section is **recommended** and does not count toward the page limit.)

It is important that the work published in ICLR is reproducible. Authors are strongly encouraged to include a paragraph-long Reproducibility Statement at the end of the main text (before references) to discuss the efforts that have been made to ensure reproducibility. This paragraph should not itself describe details needed for reproducing the results, but rather reference the parts of the main paper, appendix, and supplemental materials that will help with reproducibility. For example, for novel models or algorithms, a link to an anonymous downloadable source code can be submitted as supplementary materials; for theoretical results, clear explanations of any assumptions and a complete proof of the claims can be included in the appendix; for any datasets used in the experiments, a complete description of the data processing steps can be provided in the supplementary materials. Each of the above are examples of things that can be referenced in the reproducibility statement.

AUTHOR CONTRIBUTIONS

If you'd like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

ACKNOWLEDGMENTS

Use unnumbered third level headings for the acknowledgments. All acknowledgments, including those to funding agencies, go at the end of the paper.

REFERENCES

- Sherif El Abd, Vaibhav Aggarwal, Robin Algayres, Alek Andreev, Olivier Bachem, Ian Ballantyne, Cormac Brick, Victor Cărbune, Michelle Casbon, et al. Gemma 4 technical report. *arXiv preprint arXiv:2607.02770*, 2026.
- Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. *International Conference on Learning Representations*, 2024.
- Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. Learn then test: Calibrating predictive algorithms to achieve risk control. *The Annals of Applied Statistics*, 19(2):1641–1662, 2025. doi: 10.1214/24-AOAS1998.
- Amos Azaria and Tom Mitchell. The internal state of an llm knows when it's lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 967–976, 2023.
- David Bani-Harouni, Chantal Pellegrini, Paul Stangel, Ege Özsoy, Kamilia Zaripova, Nassir Navab, and Matthias Keicher. Rewarding doubt: A reinforcement learning approach to calibrated confidence expression of large language models. In *The Fourteenth International Conference on Learning Representations*, 2026.
- Jonathan Berant, Andrew Chou, Roy Frostig, and Percy Liang. Semantic parsing on Freebase from question-answer pairs. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*, pp. 1533–1544, 2013.
- Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.
- Jon-Paul Cacioli. Verbal confidence saturation in 3–9B open-weight instruction-tuned LLMs: A pre-registered psychometric validity screen. *arXiv preprint arXiv:2604.22215*, 2026.
- Tianzhe Chu, Yuexiang Zhai, Jihan Yang, Shengbang Tong, Saining Xie, Dale Schuurmans, Quoc V Le, Sergey Levine, and Yi Ma. SFT memorizes, RL generalizes: A comparative study of foundation model post-training. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=dYur3yabMj>.
- C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- Yuyang Dai and Yuxia Wang. Rescaling confidence: What scale design reveals about LLM metacognition. *arXiv preprint arXiv:2603.09309*, 2026.
- Mehul Damani, Isha Puri, Stewart Slocum, Idan Shenfeld, Leshem Choshen, Yoon Kim, and Jacob Andreas. Beyond binary rewards: Training LMs to reason about their uncertainty. In *International Conference on Learning Representations*, 2026.

- Jasper Dekoninck, Nikola Jovanović, Tim Gehrunger, Kári Rögvaldsson, Ivo Petrov, Chenhao Sun, and Martin Vechev. Beyond benchmarks: Matharena as an evaluation platform for mathematics with LLMs. In *3rd AI for Math Workshop: Toward Self-Evolving Scientific Agents, ICML Workshop AI4Math*, 2026.
- Maksym Del, Markus K"angsepp, Marharyta Domnich, Ardi Tampuu, Lisa Yankovskaya, Meelis Kull, and Mark Fishel. How uncertainty estimation scales with sampling in reasoning models. *arXiv preprint arXiv:2603.19118*, 2026.
- Yukun Ding, Jinglan Liu, Jinjun Xiong, and Yiyu Shi. Revisiting the evaluation of uncertainty estimation and its application to explore model complexity-uncertainty trade-off. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2020.
- Dheeru Dua, Yizhong Wang, Pradeep Dasigi, Gabriel Stanovsky, Sameer Singh, and Matt Gardner. DROP: A reading comprehension benchmark requiring discrete reasoning over paragraphs. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 2368–2378, 2019.
- Allyson Ettinger, Amanda Bertsch, Bailey Kuehl, et al. Olmo 3. *arXiv preprint arXiv:2512.13961*, 2025. URL <https://arxiv.org/abs/2512.13961>.
- Aliai Eusebi, Alexander Herzog, Xiaoyu Liang, Marie Vasek, Enrico Mariconti, and Lorenzo Cavallaro. Reading calibrated uncertainty from language model trajectories. *arXiv preprint arXiv:2605.22864*, 2026.
- Conor Finlay, Joshua Kurien, Saurabh Dash, Marzieh Fadaee, and Beyza Ermis. Caliber: Calibrating confidence before and after reasoning in language models. *arXiv preprint arXiv:2606.24281*, 2026.
- Shivam Garg, Samuele Tosatto, Yangchen Pan, Martha White, and A. Rupam Mahmood. An alternate policy gradient estimator for softmax policies, 2022.
- Yonatan Geifman, Guy Uziel, and Ran El-Yaniv. Bias-reduced uncertainty estimation for deep neural classifiers. In *International Conference on Learning Representations*, 2019.
- Jiahui Geng, Fengyu Cai, Yuxia Wang, Heinz Koepl, Preslav Nakov, and Iryna Gurevych. A survey of confidence estimation and calibration in large language models. In *Proceedings of the 2024 conference of the north American chapter of the association for computational linguistics: human language technologies (volume 1: long papers)*, pp. 6577–6595, 2024.
- Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pp. 1321–1330, 2017.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Sophia Hager, David Mueller, Kevin Duh, and Nicholas Andrews. Uncertainty distillation: Teaching language models to express semantic confidence. *arXiv preprint arXiv:2503.14749*, 2025.
- Sai Hao, Hao Zeng, Hongxin Wei, and Bingyi Jing. RACER: Risk-aware calibrated efficient routing for large language models. *arXiv preprint arXiv:2603.06616*, 2026.
- Chaoqun He, Renjie Luo, Yuzhuo Bai, Shengding Hu, Zhen Leng Thai, Junhao Shen, Jinyi Hu, Xu Han, Yujie Huang, Yuxiang Zhang, Jie Liu, Lei Qi, Zhiyuan Liu, and Maosong Sun. Olympiad-bench: A challenging benchmark for promoting AGI with olympiad-level bilingual multimodal scientific problems. *arXiv preprint arXiv:2402.14008*, 2024.

- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In *Advances in Neural Information Processing Systems*, 2021.
- Huipeng Huang, Wenbo Liao, Huajun Xi, Hao Zeng, Mengchen Zhao, and Hongxin Wei. Model-agnostic selective labeling with provable statistical guarantees. *arXiv preprint arXiv:2510.14581*, 2025.
- Xinmeng Huang, Shuo Li, Mengxin Yu, Matteo Sesia, Hamed Hassani, Insup Lee, Osbert Bastani, and Edgar Dobriban. Uncertainty in language models: Assessment through rank-calibration. *arXiv preprint arXiv:2404.03163*, 2024.
- Chaeyun Jang, Moonseok Choi, Yegon Kim, Hyungi Lee, and Juho Lee. Verbalized confidence triggers self-verification: Emergent behavior without explicit reasoning supervision. *arXiv preprint arXiv:2506.03723*, 2025. URL <https://arxiv.org/abs/2506.03723>.
- Mandar Joshi, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics*, pp. 1601–1611, 2017.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- Yuqing Kong, Mingyu Song, Yizhou Wang, and Yifan Wu. Calibration without ground truth. *arXiv preprint arXiv:2601.19862*, 2026.
- Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. *arXiv preprint arXiv:2302.09664*, 2023.
- Kenton Lee, Ming-Wei Chang, and Kristina Toutanova. Latent retrieval for weakly supervised open domain question answering. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 6086–6096, 2019. doi: 10.18653/v1/P19-1612.
- Jixuan Leng, Chengsong Huang, Banghua Zhu, and Jiaxin Huang. Taming overconfidence in LLMs: Reward calibration in RLHF. In *International Conference on Learning Representations*, 2025.
- Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, Yuhuai Wu, Behnam Neyshabur, Guy Gur-Ari, and Vedant Misra. Solving quantitative reasoning problems with language models. In *Advances in Neural Information Processing Systems*, 2022.
- Changcheng Li, Jiancan Wu, Hengheng Zhang, Zhengsu Chen, Guo An, Junxiang Qiu, Xiang Wang, and Qi Tian. Confidence before answering: A paradigm shift for efficient llm uncertainty estimation. *arXiv preprint arXiv:2603.05881*, 2026a.
- Chen Li, Xiaoling Hu, Songzhu Zheng, Jiawei Zhou, and Chao Chen. ORCE: Order-aware alignment of verbalized confidence in large language models. *arXiv preprint arXiv:2605.12446*, 2026b.
- Yibo Li, Miao Xiong, Jiaying Wu, and Bryan Hooi. ConfTuner: Training large language models to express their confidence verbally. *arXiv preprint arXiv:2508.18847*, 2025.
- Stephanie Lin, Jacob Hilton, and Owain Evans. Teaching models to express their uncertainty in words. *Transactions on Machine Learning Research*, 2022.
- Shudong Liu, Zhaocong Li, Xuebo Liu, Runzhe Zhan, Derek F. Wong, Lidia S. Chao, and Min Zhang. Can LLMs learn uncertainty on their own? expressing uncertainty effectively in a self-training manner. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 21635–21645, 2024a. doi: 10.18653/v1/2024.emnlp-main.1205.
- Xiaoou Liu, Tiejun Chen, Longchao Da, Chacha Chen, Zhen Lin, and Hua Wei. Uncertainty quantification and confidence calibration in large language models: A survey. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*, pp. 6107–6117, 2025.

- Xin Liu, Muhammad Khalifa, and Lu Wang. LitCab: Lightweight language model calibration over short- and long-form responses. In *International Conference on Learning Representations*, 2024b.
- Beier Luo, Shuoyuan Wang, Sharon Li, and Hongxin Wei. Your pre-trained llm is secretly an unsupervised confidence calibrator. *Advances in Neural Information Processing Systems*, 38: 163514–163543, 2026.
- Michael Luo, Sijun Tan, Justin Wong, Xiaoxiang Shi, William Y. Tang, Manan Roongta, Colin Cai, Jeffrey Luo, Li Erran Li, Raluca Ada Popa, and Ion Stoica. Deepscaler: Surpassing o1-preview with a 1.5b model by scaling rl, 2025. Notion Blog.
- Zhengzhao Ma, Xueru Wen, Boxi Cao, Yaojie Lu, Hongyu Lin, Jinglin Yang, Min He, Xianpei Han, and Le Sun. Decoupling reasoning and confidence: Resurrecting calibration in reinforcement learning from verifiable rewards. In *Forty-third International Conference on Machine Learning*, 2026.
- Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 9802–9822, 2023. doi: 10.18653/v1/2023.acl-long.546.
- Zhiting Mei, Christina Zhang, Tenny Yin, Justin Lidard, Ola Shorinwa, and Anirudha Majumdar. Reasoning about uncertainty: Do reasoning models know when they don’t know? *arXiv preprint arXiv:2506.18183*, 2025.
- Allan H. Murphy. A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4):595–600, 1973.
- Shiyu Ni, Keping Bi, Lulu Yu, and Jiafeng Guo. Are large language models more honest in their probabilistic or verbalized confidence? *arXiv preprint arXiv:2408.09773*, 2024.
- Ki Jung Seo, Sehun Lim, and Taeuk Kim. ADVICE: Answer-dependent verbalized confidence estimation. *arXiv preprint arXiv:2510.10913*, 2025.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Jianru Shen. Provable limits and certified deferral for verbalized uncertainty. *arXiv preprint arXiv:2608.05064*, 2026.
- Kaiwen Shi, Zheyuan Zhang, and Yanfang Ye. SAGE: Answer-conditioned uncertainty targets for verbal uncertainty alignment. *arXiv preprint arXiv:2606.11512*, 2026.
- Ponhvoan Srey, Xiaobao Wu, Cong-Duy Nguyen, Quang Minh Nguyen, Duc Anh Vu, and Anh Tuan Luu. From signals to transfer: A factorised study of probe-based uncertainty estimation in large language models. *arXiv preprint arXiv:2606.27679*, 2026.
- Chee Heng Tan, Zhuoyi Lin, Mehul Motani, and Wee Sun Lee. On the effectiveness of reward functions in reinforcement learning for confidence calibration of large language models. *arXiv preprint arXiv:2607.04332*, 2026.
- Linwei Tao, Yi-Fan Yeh, Mingjing Dong, Tao Huang, Philip Torr, and Chang Xu. Revisiting uncertainty estimation and calibration of large language models. *arXiv preprint arXiv:2505.23854*, 2025.
- Amir Taubenfeld, Tom Sheffer, Eran Ofek, Amir Feder, Ariel Goldstein, Zorik Gekhman, and Gal Yona. Confidence improves self-consistency in LLMs. *arXiv preprint arXiv:2502.06233*, 2025.
- Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D. Manning. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. *arXiv preprint arXiv:2305.14975*, 2023.

- Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. MuSiQue: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554, 2022.
- Liaoyaqi Wang, Chunsheng Zuo, William Jurayj, Benjamin Van Durme, and Anqi Liu. Process supervision of confidence margin for calibrated LLM reasoning. In *Conference on Language Modeling*, 2026.
- Victor Wang and Elias Stengel-Eskin. Calibrating verbalized confidence with self-generated distractors. In *International Conference on Learning Representations*, volume 2026, pp. 121268–121297, 2026.
- Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Ben Feuer, Siddhartha Jain, Ravid Shwartz-Ziv, Neel Jain, Khalid Saifullah, Siddhartha Naidu, Chinmay Hegde, Yann LeCun, Tom Goldstein, Willie Neiswanger, and Micah Goldblum. LiveBench: A challenging, contamination-free LLM benchmark. *arXiv preprint arXiv:2406.19314*, 2024. URL <https://arxiv.org/abs/2406.19314>.
- Jiayun Wu, Jiashuo Liu, Zhiyuan Zeng, Tianyang Zhan, Tianle Cai, and Wenhao Huang. Mitigating LLM hallucination via behaviorally calibrated reinforcement learning. In *Agent AI in the Wild: From Hallucinations to Reliable Autonomy*, 2026.
- Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs. In *International Conference on Learning Representations*, 2024.
- Tianyang Xu, Shujin Wu, Shizhe Diao, Xiaozhe Liu, Xingyao Wang, Yangyi Chen, and Jing Gao. SaySelf: Teaching LLMs to express confidence with self-reflective rationales. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 5985–5998, 2024.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- Daniel Yang, Yao-Hung Hubert Tsai, and Makoto Yamada. On verbalized confidence scores for LLMs. *arXiv preprint arXiv:2412.14737*, 2024.
- Sin-Han Yang, Cheng-Kuang Wu, Chieh-Yen Lin, Yun-Nung Chen, Hung-yi Lee, and Shao-Hua Sun. On calibration of large language models: From response to capability. *arXiv preprint arXiv:2602.13540*, 2026a.
- Xuqing Yang, Yi Yuan, Shanzhe Lei, and Xuhong Wang. Scaling with confidence: Calibrating confidence of llms for adaptive test time scaling. *arXiv preprint arXiv:2607.01612*, 2026b.
- Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pp. 2369–2380, 2018.
- Yi-Fan Yeh, Linwei Tao, Minjing Dong, Tao Huang, Jialin Yu, Philip Torr, and Chang Xu. Retrieval-augmented linguistic calibration. *arXiv preprint arXiv:2605.19344*, 2026.
- Chengyao Yu, Hao Zeng, Youxin Zhu, Jianguo Huang, Huajun Zeng, and Bingyi Jing. Anytime safe PAC efficient reasoning. *International Conference on Machine Learning*, 2026a.
- Fengfei Yu, Ruijia Niu, Dongxia Wu, Yian Ma, and Rose Yu. Calibrating LLMs with semantic-level reward. *arXiv preprint arXiv:2605.15588*, 2026b.
- Qiyang Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan, Gaohong Liu, Lingjun Liu, et al. DAPO: An open-source LLM reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.

Hao Zeng, Jianguo Huang, Bingyi Jing, Hongxin Wei, and Bo An. On the provable performance guarantee of efficient reasoning models. *arXiv preprint arXiv:2510.09133*, 2025.

Caiqi Zhang, Xiaochen Zhu, Chengzu Li, Nigel Collier, and Andreas Vlachos. LoVeC: Reinforcement learning for better verbalized confidence in long-form generation. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 33336–33363. Association for Computational Linguistics, 2026a. doi: 10.18653/v1/2026.acl-long.1539.

Chuang Zhang, Zizhen Zhu, Yihao Wei, Bing Tian, Junyi Liu, Henan Wang, Xavier Wang, and Yaxiao Liu. Confidence-calibrated small-large language model collaboration for cost-efficient reasoning. *arXiv preprint arXiv:2603.03752*, 2026b.

Hanning Zhang, Shizhe Diao, Yong Lin, Yi R. Fung, Qing Lian, Xingyao Wang, Yangyi Chen, Heng Ji, and Tong Zhang. R-Tuning: Instructing large language models to say ‘I don’t know’. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*, 2024.

APPENDIX

A Related Work	17
B Theoretical Analysis	18
B.1 Proof of Proposition 1	18
B.2 Proof of Proposition 2	19
B.3 Proof of Proposition 3	19
C Implementation Details	20
C.1 Training Pipelines	20
C.2 Evaluation Protocol	21
C.3 Confidence Formats	21
D Detailed Experimental Results	22
D.1 Per-Dataset Main Results	22
D.2 Training Dynamics	22
D.3 Stability Across Training Seeds	25
D.4 Ablation Studies	26
D.5 Results under Additional Metrics	28
D.6 Integration with Other Confidence Estimators	30
D.7 Confidence Visualization	30
D.8 Annotation and Compute Cost	32
E Downstream Applications	32
E.1 Selective Risk Control	32
E.2 Confidence-Guided Model Routing	33

A RELATED WORK

Confidence estimation in LLMs. Confidence estimates for a model answer can be derived from many signals, of which four families cover most existing methods and differ in what they require at inference time (Geng et al., 2024; Liu et al., 2025). The most direct signal is the probability the model assigns to its own answer tokens, used as the confidence either as it is or after a layer fitted on labeled data corrects it (Kadavath et al., 2022; Liu et al., 2024b). Probing methods instead train a classifier on the hidden states or on the layer-wise trajectory that produces the answer (Azaria & Mitchell, 2023; Eusebi et al., 2026; Srey et al., 2026), which requires access to activations rather than logits. When a model exposes neither, sampling-based methods recover confidence from behavior alone and measure how far several sampled answers agree (Kuhn et al., 2023; Del et al., 2026; Taubenfeld et al., 2025), at the cost of repeated generations per question. Verbalized confidence removes that cost as well: the model states a numerical or linguistic score alongside its answer within a single generation (Lin et al., 2022; Tian et al., 2023; Xiong et al., 2024; Yang et al., 2024). We study the verbalized setting for reasoning models, where the score follows a reasoning trace and the final answer. Such a score is informative only once it matches accuracy: the values an untrained model reports concentrate on a few points of the scale (Dai & Wang, 2026; Cacioli, 2026) and rank its errors worse than its own token probabilities (Ni et al., 2024; Tao et al., 2025).

Confidence calibration for LLMs. Calibration requires the reported confidence to match the empirical probability that the answer is correct, and a method either rescales a score after the model has produced it or changes the model that produces it. Rescaling is the standard treatment for probability-based signals: a temperature or a bias layer is fitted on a labeled held-out set (Guo et al., 2017; Liu et al., 2024b), and label-free variants align the score with a reference model or project the two predictive distributions onto a calibrated one (Luo et al., 2026; Kong et al., 2026). Probes need no separate step, since they are classifiers already trained on correctness labels (Azaria & Mitchell, 2023; Srey et al., 2026), and a sampling-based estimate is calibrated by choosing how agreement among samples maps to a probability (Kuhn et al., 2023; Del et al., 2026). A verbalized score can be rescaled the same way, but only through extra queries per question or an extra pass that rewrites the reported value (Wang & Stengel-Eskin, 2026; Yeh et al., 2026), and neither changes how the model generates the score in the first place.

Post-training instead changes the model that emits the score, and it targets the verbalized score because that score needs no logit access and no extra samples at deployment. Supervised fine-tuning supervises the score with correctness signals collected from the model’s own responses (Lin et al., 2022; Zhang et al., 2024; Jang et al., 2025; Liu et al., 2024a; Xu et al., 2024; Hager et al., 2025; Seo et al., 2025; Zhang et al., 2026a; Li et al., 2025). Reinforcement learning adds a calibration term to the reward of RLVR, which by itself optimizes answer correctness and leaves the confidence unsupervised (Shao et al., 2024; Yu et al., 2025); the term is a proper scoring rule on the reported confidence, and the methods differ in where the score is placed, what it is trained toward, and how credit reaches its tokens (Leng et al., 2025; Damani et al., 2026; Bani-Harouni et al., 2026; Tan et al., 2026; Li et al., 2026a; Finlay et al., 2026; Shi et al., 2026; Yu et al., 2026b; Li et al., 2026b; Ma et al., 2026; Wang et al., 2026; Yang et al., 2026b; Zhang et al., 2026b). These objectives learn from the confidence values the base model already samples, which Section 3 shows are concentrated on a few high values and stay concentrated while the calibration reward is optimized. In this paper, we analyze this prior and propose CalibSFT, an SFT stage that initializes verbalized confidence before confidence-aware RL.

B THEORETICAL ANALYSIS

B.1 PROOF OF PROPOSITION 1

Proof. We fix x and the expected rewards u_i , and differentiate with respect to the confidence logits. The softmax derivative is $\partial\pi_j/\partial\eta_i = \pi_j(\mathbf{1}\{j = i\} - \pi_i)$. It follows that

$$\frac{\partial J_x}{\partial\eta_i} = \sum_j u_j \pi_j (\mathbf{1}\{j = i\} - \pi_i) \quad (11)$$

$$= \pi_i(u_i - J_x). \quad (12)$$

Since J_x is a convex combination of the u_i values, $|u_i - J_x| \leq \Delta_x$, which gives the coordinate-wise bound. If $\pi_k = 1 - \varepsilon$, then

$$|u_k - J_x| = \left| \sum_{i \neq k} \pi_i (u_k - u_i) \right| \leq \varepsilon \Delta_x. \quad (13)$$

Consequently,

$$\|\nabla_\eta J_x\|_1 = \pi_k |u_k - J_x| + \sum_{i \neq k} \pi_i |u_i - J_x| \quad (14)$$

$$\leq (1 - \varepsilon)\varepsilon\Delta_x + \varepsilon\Delta_x \quad (15)$$

$$\leq 2\varepsilon\Delta_x. \quad (16)$$

□

The gradient is the expectation of the likelihood-ratio estimator $U\nabla_\eta \log \pi_I$, where $I \sim \pi$ and $U = -(c_I - Z)^2$, with Z drawn from its fixed conditional distribution given x , independently of I . This matches RLVR’s reward $Z - (c_I - Z)^2$ (Preliminaries) up to the term Z , which does not

depend on c_I . Assuming $C \perp Z \mid x$ as in Proposition 2, this term contributes an η -independent additive constant to J_x that does not affect this gradient. A baseline depending only on x leaves this expectation unchanged.

B.2 PROOF OF PROPOSITION 2

Proof. Conditional independence gives $\mathbb{E}[Z - p_x \mid x, C] = 0$. Expanding the square therefore yields

$$\mathbb{E}[(C - Z)^2 \mid x] = \mathbb{E}[(Z - p_x)^2 \mid x] + \mathbb{E}[(C - p_x)^2 \mid x] \quad (17)$$

$$= p_x(1 - p_x) + \mathbb{E}[(C - p_x)^2 \mid x]. \quad (18)$$

For any $\delta > 0$, the second term satisfies

$$\mathbb{E}[(C - Z)^2 \mid x] - p_x(1 - p_x) = \mathbb{E}[(C - p_x)^2 \mid x] \quad (19)$$

$$\geq \delta^2 \Pr(|C - p_x| \geq \delta \mid x) \quad (20)$$

$$= \delta^2(1 - m_\delta(x)). \quad (21)$$

Thus, assigning little probability mass near p_x incurs excess Brier loss even when confidence values in that neighborhood remain possible. $\square$

Restricted confidence support. As a special case, suppose $\Pr(C \in S_x \mid x) = 1$ for a nonempty set $S_x \subseteq [0, 1]$, and define $d_x = \text{dist}(p_x, S_x) = \inf_{c \in S_x} |c - p_x|$. If $d_x > 0$, then $m_{d_x}(x) = 0$, and Proposition 2 with $\delta = d_x$ gives

$$\mathbb{E}[(C - Z)^2 \mid x] \geq p_x(1 - p_x) + \text{dist}(p_x, S_x)^2. \quad (22)$$

For $d_x = 0$, the same bound follows directly from the decomposition above. If S_x is closed, a predictor that always outputs a nearest point in S_x attains the bound. Confidence values are effectively drawn from a small discrete set throughout this paper (Figure 1), so $\text{dist}(p_x, S_x)$ measures how far the nearest available value falls from p_x , translating the observed concentration directly into excess Brier risk.

Best-of- G sampling under a concentrated policy. Given the policy and a question x , draw G independent confidence samples $C_1, \dots, C_G$ from its conditional distribution. Each sample lies outside $\{c : |c - p_x| < \delta\}$ with probability $1 - m_\delta(x)$, and by independence all G samples do so with probability $(1 - m_\delta(x))^G$. On that event, even the sample closest to p_x has squared error at least δ^2 , so

$$\mathbb{E} \left[\min_{i \leq G} (C_i - p_x)^2 \mid x \right] \geq \delta^2 (1 - m_\delta(x))^G. \quad (23)$$

With the group size $G = 8$ used in our RL runs, a policy that rarely samples near p_x is unlikely to produce a well-calibrated confidence anywhere in the group.

B.3 PROOF OF PROPOSITION 3

We analyze the unrounded targets under the original sampling distribution, before response filtering and target balancing. The random variables Q_x and C_i^* correspond to the observed q_x and c_i^* in Eq. 8.

Proof of Proposition 3. Fix x and write $v_x = p_x(1 - p_x)$. Conditional independence gives

$$\mathbb{E}[Q_x \mid x] = \mathbb{E}[Z_i \mid x] = p_x, \quad \text{Var}(Q_x \mid x) = \frac{v_x}{n}. \quad (24)$$

Since Z_i is included in Q_x , their conditional covariance is

$$\text{Cov}(Q_x, Z_i \mid x) = \frac{v_x}{n}. \quad (25)$$

Linearity of expectation gives $\mathbb{E}[C_i^* \mid x] = p_x$. Expanding the variance yields

$$\text{Var}(C_i^* \mid x) = \lambda^2 \frac{v_x}{n} + (1 - \lambda)^2 v_x + 2\lambda(1 - \lambda) \frac{v_x}{n} \quad (26)$$

$$= v_x \left[\frac{1}{n} + (1 - \lambda)^2 \left(1 - \frac{1}{n} \right) \right]. \quad (27)$$

Now fix a at which ℓ is finite. Since $\ell(a, t)$ is affine in t , unbiasedness gives

$$\mathbb{E}[\ell(a, C_i^*) \mid x] = p_x \ell(a, 1) + (1 - p_x) \ell(a, 0) = \mathbb{E}[\ell(a, Z_i) \mid x]. \quad (28)$$

Because ℓ is strictly proper, this expected loss is uniquely minimized at $a = p_x$. Thus the mixture, the group average, and an individual correctness label induce the same expected scoring loss for a prediction based on x . For the Brier loss $\ell(a, z) = (a - z)^2$, the extension is $\ell(a, t) = (a - t)^2 + t(1 - t)$, so the squared loss against a soft target has the same minimizer. The same argument applies to the binary logarithmic loss (Gneiting & Raftery, 2007). $\square$

Effect of the response count and mixing weight. At $\lambda = 0$, the target is the binary label with variance v_x ; at $\lambda = 1$, it is the group average with variance v_x/n . For a fixed $\lambda < 1$, the variance approaches $(1 - \lambda)^2 v_x$ as n increases, reflecting the retained contribution of the individual outcome. Within any observed group, $C_i^* - C_j^* = (1 - \lambda)(Z_i - Z_j)$, so the targets of a correct and an incorrect response differ by $1 - \lambda$.

Conditioning on the generated response. Proposition 3 concerns the success probability p_x before observing a particular response. To make the effect of conditioning explicit, let h_i include x and information from response i , and define $p_{h_i} = \Pr(Z_i = 1 \mid h_i)$. If the other responses are independent of h_i conditional on x , then

$$\mathbb{E}[C_i^* \mid h_i] = \left(1 - \lambda + \frac{\lambda}{n}\right) p_{h_i} + \lambda \left(1 - \frac{1}{n}\right) p_x. \quad (29)$$

The target thus shrinks the response-conditional probability toward the question’s success probability. These results justify the mean and variance of this supervision signal under the original sampling distribution; it does not establish response-conditional calibration after target balancing and token-level SFT. Those effects are evaluated empirically.

C IMPLEMENTATION DETAILS

C.1 TRAINING PIPELINES

All runs use Qwen3-8B as the base model. From DeepScaleR-Preview we remove duplicate questions and questions whose gold answers disagree across duplicates, which leaves 38,915 questions. We randomly hold out 2,000 of them as a test split (DeepScaleR-Eval) and use the remaining 36,915 as the training split (DeepScaleR-Train), which is used to construct the SFT data for CalibSFT and the downstream RL training data.

SFT data construction. We sample $n = 50$ responses per question from DeepScaleR-Train with the base model, using temperature 1.0, top- p 0.95, and a cap of 8,192 tokens, which gives 1,845,750 responses in total. Each question’s success rate q_x is computed from all 50 responses before any filtering. We then discard responses that do not follow the required `<think><answer><confidence>` format and responses that hit the token cap without terminating, which are typically repetitive or looping generations. This removes 401,634 responses and retains 1,444,116, with q_x kept fixed. For each retained response we form the target $c_i^* = \lambda q_x + (1 - \lambda) z_i$ with $\lambda = 0.5$ and replace the confidence suffix with it, preserving the sampled reasoning and answer. In the main probability format, the target is rendered to two decimal places in the trailing `<confidence>` tag, which yields 100 distinct target values. For balanced sampling, we group examples by the numeric target rounded to six decimal places and sample the same number from every nonempty group, using the smallest group size as the default count. We apply this rule either through a per-epoch sampler or when constructing an offline balanced split; the offline split used in our main experiments contains 349 examples per target value, that is, 34,900 examples drawn from 11,539 distinct questions.

CalibSFT. We train on the balanced split with the correctness-conditional loss mask and $\beta = 0.5$: correct responses supervise both the reasoning/answer and the confidence segments, whereas incorrect responses supervise only the confidence segment. We use AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with no weight decay and gradient-norm clipping at 1.0, a constant learning rate of 2×10^{-6} after 10 warm-up steps, and a global batch size of 256 sequences truncated to 16,384 tokens. We train for 180 optimization steps, roughly 1.3 epochs over the balanced split, on eight B20Z GPUs.

Confidence-aware RL. For downstream RL, each prompt group contains 8 sampled responses, with each response capped at 8,192 tokens. We train with batches of 64 prompt groups for 160 optimization steps. For methods using DAPO dynamic sampling, this corresponds to approximately 600 generation batches. Dynamic sampling retains only groups with mixed correctness outcomes and discards groups whose responses are all correct or all incorrect. To accelerate training for these methods, we remove questions for which all sampled responses from the base model are correct before RL training begins. Each method retains its own reward and credit assignment.

- **Direct RL baselines.** RLVR, RLCR (Damani et al., 2026), CoCA (Li et al., 2026a), and DCPO (Ma et al., 2026) are trained directly from the Qwen3-8B base checkpoint. They share the DAPO training recipe but retain their respective correctness, calibration-target, and credit-assignment objectives.
- **ReDoubt.** ReDoubt is initialized from the trained RLVR checkpoint and then optimized with its clipped log-score confidence objective (Bani-Harouni et al., 2026). When incorporated with CalibSFT, we first train CalibSFT on the RLVR checkpoint and then continue with ReDoubt RL.
- **C3RL.** C3RL (Yang et al., 2026b) combines correctness, calibration, and reference-accuracy rewards. C3RL uses the digit confidence format, expressing confidence as an integer from 0 to 9 and mapping it to $[0, 1]$ by dividing by 9. The CalibSFT initialization for C3RL is also trained in this format. We use the base model’s success rates over sampled responses to label each training question as all correct, partially correct, or all incorrect. The reference reward encourages correct answers to previously all-incorrect questions and penalizes incorrect answers to previously all-correct questions. To retain these questions and their reference-reward signals, we disable DAPO dynamic sampling and all-correct filtering, using the full RL training split for C3RL both with and without CalibSFT initialization.

C.2 EVALUATION PROTOCOL

For the main results, we use temperature 0.6 and sample four responses per question on non-AIME mathematical and OOD benchmarks and 32 responses per question on AIME. Pass@1 is the mean correctness over these sampled responses. The supervision and target ablations retain their recorded evaluation settings. Confidence is parsed from the final probability field and normalized to $[0, 1]$. We use 10 equal-width bins for calculating ECE. Unless otherwise noted, RL results use the final checkpoint after 160 optimization steps, while SFT-only results use the checkpoint at the end of their respective SFT runs.

Answer verification. Every benchmark is scored by a deterministic verifier except TriviaQA. On the mathematical benchmarks and HotpotQA, a response is correct if its extracted answer is symbolically equivalent to the gold answer under `math_verify`, or if the two agree after lowercasing, removing punctuation and articles, and collapsing whitespace. NQOpen and MuSiQue apply the same normalized match against any gold alias. PopQA follows its official protocol and accepts a response that contains a gold alias, and WebQuestions requires the set of predicted answer items to match the gold set exactly. DROP uses the official DROP normalization and counts a response correct only on an exact match with one of its references. LiveBenchReasoning uses LiveBench’s own deterministic graders for its spatial, web-of-lies, and zebra-puzzle tasks. TriviaQA admits many surface forms per answer, so we score it with an LLM judge: Qwen3-8B receives the question, the reference answer, and the candidate answer, and returns a correct or incorrect verdict. The same symbolic-equivalence-or-normalized-match rule produces the correctness labels z_i that build the CalibSFT targets and the correctness rewards during RL.

C.3 CONFIDENCE FORMATS

The format comparison in Section 5.4 keeps the CalibSFT training data fixed and changes only how the confidence targets are written. Probability targets use two decimal places, and we compare two target grids with steps of 0.01 and 0.05. Integer targets map a target c^* to $\lceil 9c^* + 0.5 \rceil \in \{0, \dots, 9\}$, and linguistic targets assign *low*, *medium*, and *high* to $c^* < 1/3$, $1/3 \leq c^* < 2/3$, and $c^* \geq 2/3$, respectively. Table 5 lists the system prompt used for each format. For evaluation, an integer d is mapped to $d/9$ and the labels *low*, *medium*, and *high* to 0.25, 0.5, and 0.75.

Table 5: System prompts for the three confidence formats.

Format	System prompt
Probability	A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and analyzes its confidence about the solution and then provides the user with the final answer as well as its confidence level. The confidence level indicates how certain the Assistant is about its answer, expressed as a decimal between 0 and 1 with exactly two decimal places, enclosed within <code><confidence></code> <code></confidence></code> tags. The response must strictly follow this format: <code><think></code> reasoning process here <code></think></code> <code><answer></code> final short answer only <code></answer></code> <code><confidence></code> 0.xx <code></confidence></code> . The <code><answer></code> tag must contain only the final answer string needed for exact-match evaluation, not a full sentence, explanation, or reasoning.
Integer	A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and analyzes its confidence about the solution and then provides the user with the final answer as well as its confidence score. The confidence score indicates how certain the Assistant is about its answer, expressed as a single integer from 0 to 9, enclosed within <code><confidence></code> <code></confidence></code> tags. A score of 0 means very low confidence and the answer is likely incorrect. A score of 9 means very high confidence and the answer is very likely correct. The response must strictly follow this format: <code><think></code> reasoning process here <code></think></code> <code><answer></code> final short answer only <code></answer></code> <code><confidence></code> X <code></confidence></code> , where X is one of 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. The <code><answer></code> tag must contain only the final answer string needed for exact-match evaluation, not a full sentence, explanation, or reasoning.
Linguistic	A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and analyzes its confidence about the solution and then provides the user with the final answer as well as its confidence level. The confidence level indicates how certain the Assistant is about its answer, expressed as exactly one of low, medium, or high, enclosed within <code><confidence></code> <code></confidence></code> tags. Low means the answer is likely incorrect, medium means the answer is uncertain, and high means the answer is likely correct. The response must strictly follow this format: <code><think></code> reasoning process here <code></think></code> <code><answer></code> final short answer only <code></answer></code> <code><confidence></code> label <code></confidence></code> . The <code><answer></code> tag must contain only the final answer string needed for exact-match evaluation, not a full sentence, explanation, or reasoning.

D DETAILED EXPERIMENTAL RESULTS

D.1 PER-DATASET MAIN RESULTS

Tables 6 and 7 report the results on each individual benchmark, complementing the split-level averages in Table 1. We present the average over its eight datasets in the last subtable of each split.

The improvements hold across benchmarks. The gains are largest where the uninitialized RL models are most overconfident. On the six benchmarks with baseline ECE above 25 (MinervaMath, MuSiQue, LiveBenchReasoning, NQOpen, PopQA, and WebQuestions), CalibSFT initialization reduces both Brier score and ECE for all methods. The Base, RLVR, and CalibSFT-alone rows show what each stage contributes. RLVR raises Pass@1 but leaves calibration unchanged, and on the out-of-distribution split it even worsens Brier score and ECE relative to Base, matching the failure mode in Section 3. CalibSFT alone substantially reduces Brier score and ECE on both splits and improves in-distribution AUROC. Confidence-aware RL then lifts out-of-distribution AUROC from 61.38 to above 71 for four of the five methods. CalibSFT sets the scale of confidence, and RL sharpens its ranking outside the training domain.

D.2 TRAINING DYNAMICS

In Section 3, we observe that confidence stays concentrated on a few values while RLCR optimizes its calibration reward. To verify that this behavior is not specific to one objective, we further train Qwen3-8B with CoCA (Li et al., 2026a) and DCPO (Ma et al., 2026) under the same setup and track

Table 6: Per-dataset results on the in-domain mathematical benchmarks. Base, RLVR, and standalone CalibSFT use 6,144 tokens; downstream RL models use 16,384 tokens. Temperature is 0.6, with four responses per non-AIME question and 32 per AIME question.

(a) DeepScaleR-Eval					(b) MATH-500					(c) MinervaMath				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	61.00	66.38	33.94	33.90	Base	84.50	65.22	14.18	13.27	Base	47.52	60.84	46.83	47.65
RLVR	70.26	67.94	26.85	26.26	RLVR	89.50	67.63	9.80	8.41	RLVR	52.85	60.81	43.34	43.51
CalibSFT	61.30	76.39	22.65	17.28	CalibSFT	84.00	81.51	16.49	14.06	CalibSFT	44.94	66.31	25.37	17.42
RLCR	71.47	75.45	16.56	5.02	RLCR	90.80	76.45	7.42	7.53	RLCR	51.75	61.57	29.17	23.05
+ CalibSFT	72.47	79.19	15.43	4.63	+ CalibSFT	90.35	89.86	6.46	3.89	+ CalibSFT	51.93	73.00	22.59	12.00
ReDoubt	71.62	75.91	17.12	10.28	ReDoubt	90.10	75.41	9.79	15.06	ReDoubt	52.30	60.64	25.36	10.67
+ CalibSFT	72.42	77.98	16.07	6.71	+ CalibSFT	89.70	89.44	9.59	16.99	+ CalibSFT	51.38	74.47	20.96	9.35
CoCA	70.33	73.21	17.69	8.50	CoCA	90.10	73.02	10.57	18.22	CoCA	49.63	64.28	26.82	18.83
+ CalibSFT	71.54	78.46	16.45	7.76	+ CalibSFT	90.05	87.05	9.08	15.69	+ CalibSFT	50.55	76.38	21.78	13.77
DCPO	71.66	71.72	17.23	6.36	DCPO	89.65	70.59	7.67	6.52	DCPO	50.64	63.57	31.72	29.05
+ CalibSFT	70.54	77.54	16.16	5.87	+ CalibSFT	90.40	84.55	7.71	7.83	+ CalibSFT	50.55	76.00	20.53	8.83
C3RL	70.84	62.79	24.50	24.86	C3RL	89.85	60.68	8.97	9.30	C3RL	50.37	57.50	45.46	46.34
+ CalibSFT	71.25	71.52	20.08	14.85	+ CalibSFT	88.05	84.35	10.47	9.71	+ CalibSFT	51.01	73.31	21.79	13.06
(d) OlympiadBench					(e) GSM8K					(f) AIME2024				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	61.42	74.08	32.51	33.33	Base	94.39	67.91	5.16	3.35	Base	48.23	81.93	39.61	42.50
RLVR	61.39	73.08	34.64	34.90	RLVR	94.71	68.09	4.99	2.98	RLVR	52.92	82.05	39.49	41.15
CalibSFT	60.50	85.75	16.72	11.61	CalibSFT	94.60	66.46	25.93	30.61	CalibSFT	47.08	89.04	10.35	7.73
RLCR	63.17	83.46	16.73	8.48	RLCR	94.86	78.70	4.75	5.64	RLCR	59.48	93.59	13.17	19.70
+ CalibSFT	63.28	89.00	14.60	9.25	+ CalibSFT	94.77	82.24	4.65	2.81	+ CalibSFT	64.17	92.42	10.97	9.79
ReDoubt	62.80	81.10	17.14	7.73	ReDoubt	95.00	78.24	7.07	16.31	ReDoubt	55.62	91.94	15.40	22.77
+ CalibSFT	63.58	90.10	14.29	14.69	+ CalibSFT	94.77	80.01	7.48	17.69	+ CalibSFT	61.56	93.81	11.89	14.37
CoCA	61.57	81.47	18.37	14.56	CoCA	94.66	68.97	8.91	21.10	CoCA	53.85	92.22	16.27	21.37
+ CalibSFT	61.57	88.17	17.05	17.07	+ CalibSFT	94.64	71.42	8.25	18.18	+ CalibSFT	58.54	95.41	16.34	23.56
DCPO	62.24	79.80	18.80	12.97	DCPO	94.58	68.17	5.44	9.17	DCPO	57.92	88.84	12.32	8.47
+ CalibSFT	62.91	87.43	14.76	8.93	+ CalibSFT	94.50	73.10	5.82	7.54	+ CalibSFT	58.33	93.89	11.02	11.85
C3RL	63.06	64.34	30.64	32.22	C3RL	94.43	57.33	5.28	5.18	C3RL	59.69	80.93	25.25	30.19
+ CalibSFT	63.61	80.73	18.05	15.26	+ CalibSFT	94.37	64.41	5.30	5.16	+ CalibSFT	64.27	85.24	14.04	12.99
(g) AIME2025					(h) AIME2026					(i) Overall (ID)				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	35.00	82.75	48.25	52.81	Base	42.50	76.08	45.13	48.07	Base	59.32	71.90	33.20	34.36
RLVR	43.44	84.95	45.87	49.00	RLVR	48.23	80.70	43.82	45.92	RLVR	64.16	73.16	31.10	31.52
CalibSFT	35.83	86.49	9.61	6.59	CalibSFT	39.79	87.44	11.26	8.52	CalibSFT	58.51	79.93	17.30	14.23
RLCR	43.33	95.53	13.60	23.06	RLCR	47.81	94.10	14.08	23.19	RLCR	65.33	82.36	14.43	14.46
+ CalibSFT	48.02	92.40	13.87	14.28	+ CalibSFT	52.60	92.48	13.57	12.04	+ CalibSFT	67.20	86.32	12.77	8.59
ReDoubt	44.90	94.22	15.19	26.51	ReDoubt	52.40	90.93	16.78	26.20	ReDoubt	65.59	81.05	15.48	16.94
+ CalibSFT	51.77	96.32	11.28	16.79	+ CalibSFT	55.52	93.82	11.66	17.69	+ CalibSFT	67.59	86.99	12.90	14.28
CoCA	40.31	95.39	17.04	29.31	CoCA	48.54	96.74	14.84	27.72	CoCA	63.62	80.66	16.31	19.95
+ CalibSFT	45.00	94.21	19.99	26.43	+ CalibSFT	48.65	93.96	19.52	25.63	+ CalibSFT	65.07	85.63	16.06	18.51
DCPO	41.15	91.19	13.33	14.30	DCPO	49.58	91.78	11.04	10.41	DCPO	64.68	78.21	14.69	12.16
+ CalibSFT	41.25	95.07	13.81	17.73	+ CalibSFT	50.21	93.07	12.87	14.52	+ CalibSFT	64.84	85.08	12.83	10.39
C3RL	43.96	79.01	36.25	42.69	C3RL	52.81	84.29	27.21	33.72	C3RL	65.62	68.36	25.44	28.06
+ CalibSFT	43.12	86.74	21.86	25.49	+ CalibSFT	53.65	91.31	15.69	18.30	+ CalibSFT	66.17	79.70	15.91	14.35

the same quantities. We also initialize all three objectives from CalibSFT and compare the same quantities under both initializations.

As shown in Figures 6 and 7, we observe the same phenomenon as in Section 3. Between the first and last ten steps, Brier loss falls for RLCR (26.6 to 22.4) and CoCA (25.5 to 22.1) and stays nearly flat for DCPO (24.6 to 24.1), and ECE falls in all three runs (25.8 to 8.2, 24.8 to 7.8, and 23.8 to 14.2), so each objective either improves calibration or holds it steady. Discrimination moves the other way, with AUROC falling in every run (70.8 to 66.9, 72.2 to 68.0, and 73.5 to 64.1). Confidence support does not follow. The mean number of distinct confidence values within a group of eight responses never exceeds 5.3 at any step of the three base-initialized runs, and confidence token entropy ends

Table 7: Per-dataset results on the out-of-domain benchmarks. The evaluation protocol matches Table 6.

(a) HotpotQA					(b) TriviaQA					(c) DROP				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	64.68	57.68	32.82	32.16	Base	55.24	75.80	35.68	36.93	Base	80.36	59.13	18.22	17.03
RLVR	64.58	54.16	33.10	32.17	RLVR	54.97	76.01	40.59	41.40	RLVR	83.53	59.32	15.34	13.31
CalibSFT	66.10	55.08	30.84	22.43	CalibSFT	55.53	64.10	26.74	15.17	CalibSFT	79.76	58.30	33.30	32.64
RLCR	65.80	59.10	25.97	19.77	RLCR	55.24	81.66	23.84	23.30	RLCR	81.44	63.34	14.72	5.43
+ CalibSFT	66.27	59.23	24.25	10.85	+ CalibSFT	55.25	81.53	18.89	10.71	+ CalibSFT	82.32	74.44	14.99	10.75
ReDoubt	64.83	60.64	23.52	12.11	ReDoubt	55.69	82.96	18.67	13.58	ReDoubt	83.63	63.27	14.76	10.80
+ CalibSFT	64.68	63.08	21.70	1.37	+ CalibSFT	56.14	83.92	19.27	15.58	+ CalibSFT	82.53	75.90	15.60	17.36
CoCA	64.90	56.43	22.77	6.02	CoCA	54.77	77.62	21.63	12.97	CoCA	82.47	61.58	15.03	11.56
+ CalibSFT	66.10	61.57	22.92	11.39	+ CalibSFT	54.89	81.85	20.98	13.16	+ CalibSFT	82.50	73.91	17.95	21.99
DCPO	66.15	57.88	25.33	18.77	DCPO	54.55	79.86	25.93	23.91	DCPO	83.45	59.34	13.25	2.30
+ CalibSFT	65.97	60.70	24.43	14.92	+ CalibSFT	55.12	82.56	17.59	8.25	+ CalibSFT	80.42	75.92	16.54	13.64
C3RL	65.38	52.47	33.39	33.12	C3RL	53.76	73.01	35.61	38.53	C3RL	82.65	54.56	16.37	16.07
+ CalibSFT	66.50	58.94	28.45	19.78	+ CalibSFT	54.11	71.36	22.75	12.50	+ CalibSFT	77.44	64.78	23.65	22.03
(d) MuSiQue					(e) LiveBenchReasoning					(f) NQOpen				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	45.84	62.90	41.91	42.33	Base	52.88	61.81	40.53	40.59	Base	21.12	66.31	66.04	70.59
RLVR	47.18	58.12	46.19	46.75	RLVR	29.25	60.32	61.75	64.59	RLVR	24.13	67.07	68.92	71.55
CalibSFT	48.09	58.78	28.43	16.30	CalibSFT	55.75	69.92	34.64	35.20	CalibSFT	21.57	59.58	25.69	26.36
RLCR	47.77	66.21	25.68	15.35	RLCR	31.75	68.00	28.49	29.09	RLCR	23.32	70.48	43.84	52.43
+ CalibSFT	49.97	67.87	23.22	6.13	+ CalibSFT	42.25	68.41	21.94	7.75	+ CalibSFT	23.80	72.24	26.44	30.89
ReDoubt	46.90	66.37	24.07	11.21	ReDoubt	25.75	68.44	23.51	24.86	ReDoubt	24.67	70.28	32.15	39.19
+ CalibSFT	48.52	70.04	22.42	5.52	+ CalibSFT	42.25	80.49	19.95	12.38	+ CalibSFT	24.10	70.60	26.80	31.99
CoCA	46.69	67.03	23.69	10.18	CoCA	32.12	66.75	25.79	23.64	CoCA	22.76	66.59	35.28	43.26
+ CalibSFT	48.76	66.88	23.56	7.72	+ CalibSFT	41.50	68.03	23.24	7.96	+ CalibSFT	23.72	70.77	24.86	28.69
DCPO	47.34	65.48	28.28	20.53	DCPO	31.13	67.14	32.10	33.12	DCPO	23.69	69.80	46.27	53.69
+ CalibSFT	48.96	70.49	22.75	8.55	+ CalibSFT	37.75	71.77	21.12	8.83	+ CalibSFT	23.58	73.13	22.65	25.97
C3RL	45.55	63.39	39.88	40.04	C3RL	33.00	57.18	53.55	55.46	C3RL	22.43	64.77	63.63	68.49
+ CalibSFT	47.47	66.39	24.46	11.87	+ CalibSFT	44.62	53.33	28.11	22.78	+ CalibSFT	22.08	61.18	22.25	17.54
(g) PopQA					(h) WebQuestions					(i) Overall (OOD)				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	20.85	74.94	57.89	64.10	Base	21.88	67.10	68.16	71.80	Base	45.36	65.71	45.15	46.94
RLVR	21.37	72.78	68.14	71.96	RLVR	26.02	65.76	68.29	70.41	RLVR	43.88	64.19	50.29	51.52
CalibSFT	21.06	64.33	24.14	25.70	CalibSFT	23.11	59.54	29.45	30.13	CalibSFT	46.37	61.20	29.15	25.49
RLCR	21.29	76.88	37.71	47.90	RLCR	25.07	69.40	48.86	56.11	RLCR	43.96	69.38	31.14	31.17
+ CalibSFT	21.16	76.55	24.76	32.22	+ CalibSFT	24.63	70.20	31.41	36.33	+ CalibSFT	45.71	71.31	23.24	18.20
ReDoubt	21.37	76.83	26.82	36.23	ReDoubt	26.84	70.22	37.08	43.81	ReDoubt	43.71	69.88	25.07	23.98
+ CalibSFT	21.25	77.87	24.54	32.32	+ CalibSFT	24.91	69.57	29.11	34.59	+ CalibSFT	45.55	73.93	22.42	18.89
CoCA	21.06	73.92	30.84	39.88	CoCA	24.65	64.40	38.35	45.46	CoCA	43.68	66.79	26.67	24.12
+ CalibSFT	20.88	76.99	24.12	31.76	+ CalibSFT	26.06	69.71	25.93	28.47	+ CalibSFT	45.55	71.22	22.94	18.89
DCPO	20.99	77.02	41.10	49.96	DCPO	24.59	68.48	50.52	57.04	DCPO	43.99	68.13	32.85	32.41
+ CalibSFT	21.18	76.52	21.29	28.02	+ CalibSFT	25.61	70.87	27.45	30.32	+ CalibSFT	44.82	72.75	21.73	17.31
C3RL	20.81	71.98	54.60	61.92	C3RL	23.94	60.19	67.27	70.35	C3RL	43.44	62.19	45.54	48.00
+ CalibSFT	20.66	66.98	16.56	12.95	+ CalibSFT	23.94	61.29	24.87	19.55	+ CalibSFT	44.60	63.03	23.89	17.38

below where it started in all three runs. Therefore, none of the three objectives moves the model away from sampling a few confidence values per question, which extends the observation reported for RLCR in Section 3 to all three objectives.

Initializing from CalibSFT broadens confidence exploration throughout training. Averaged over training steps, the number of distinct confidence values per question rises from 3.3–3.9 to 6.3–7.1, and confidence token entropy rises from 0.16–0.27 to 0.61–0.90. This gap holds at every step: each CalibSFT-initialized run samples at least 5.39 distinct values per question, above the maximum of 5.25 reached by any base-initialized run. Brier loss of each CalibSFT-initialized run stays below that of its base-initialized counterpart for 90% to 97% of training steps in eleven-step means, and AUROC stays above its counterpart for 91% to 100% of steps, raising its average over training by 3.4 points

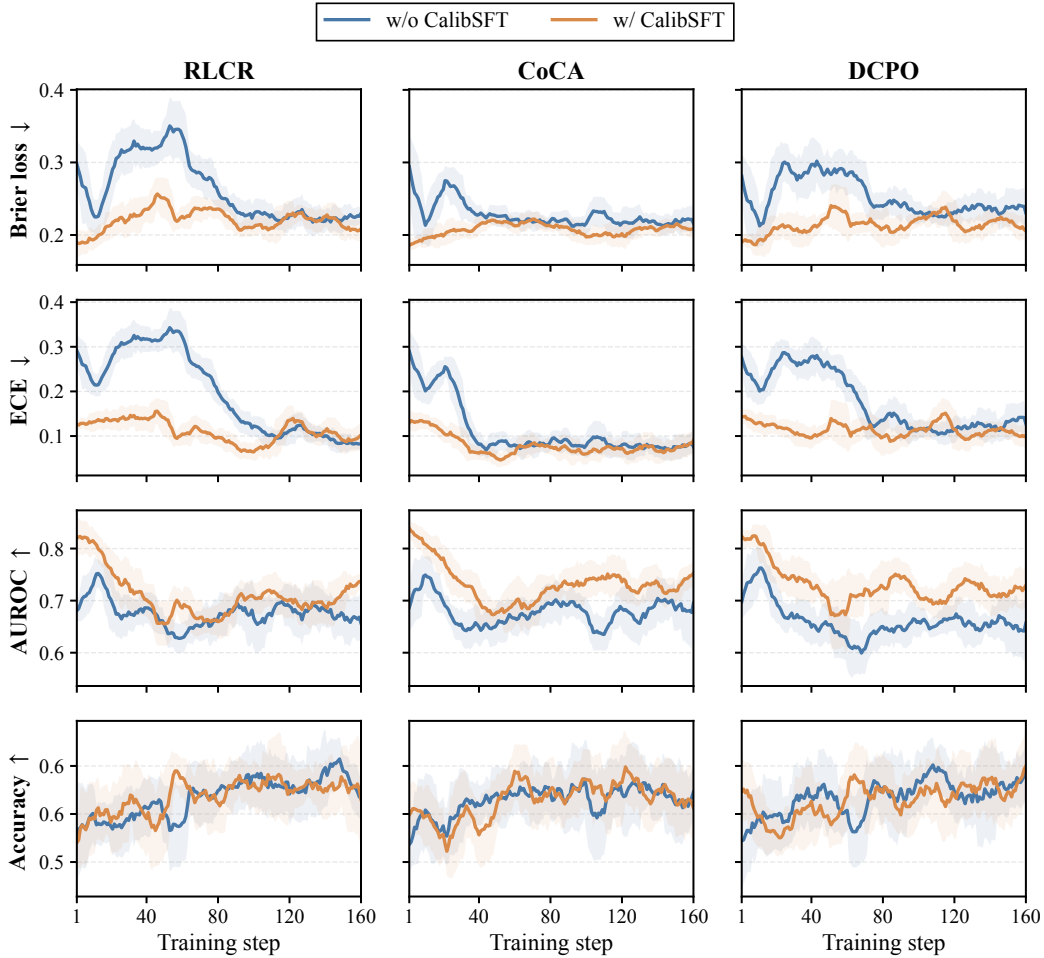


Figure 6: Brier loss, ECE, AUROC, and accuracy during training of RLCR, CoCA, and DCPO with and without CalibSFT initialization. Curves are centered eleven-step means and bands one standard deviation within each window. Statistics use the groups retained by the corresponding training run, so the sampled questions and responses change across steps.

for RLCR, 5.4 for CoCA, and 6.9 for DCPO. Accuracy remains nearly unchanged, with its average over training differing by at most 0.2 points.

D.3 STABILITY ACROSS TRAINING SEEDS

The main results report one run per method. To check that the gains from CalibSFT exceed the run-to-run variation of training, we repeat RLCR on Qwen3-8B twice per arm and evaluate every run under the protocol of Table 1. The two arms repeat different stages: the RLCR rows vary the RL seed, whereas the CalibSFT rows vary the CalibSFT seed and keep the RL seed fixed, which isolates the stage we introduce while leaving the shared RL stage unchanged.

Table 8 reports the split averages as the mean over runs with the standard deviation as a subscript. Every standard deviation is at most 1.15 points except in-distribution ECE, whose 2.51 points reflect the two CalibSFT runs reaching 8.59 and 12.13. Comparing the two arms run by run, the ranges do not overlap on any calibration or discrimination metric: the worst CalibSFT run still reaches an AUROC of 86.05 in-distribution and 71.31 out-of-distribution against best RLCR runs of 82.36 and 69.38, and its out-of-distribution ECE of 18.20 remains 12.97 points below the best RLCR run. In-distribution Pass@1 is the one metric whose ranges overlap, 65.33–66.27 for RLCR against

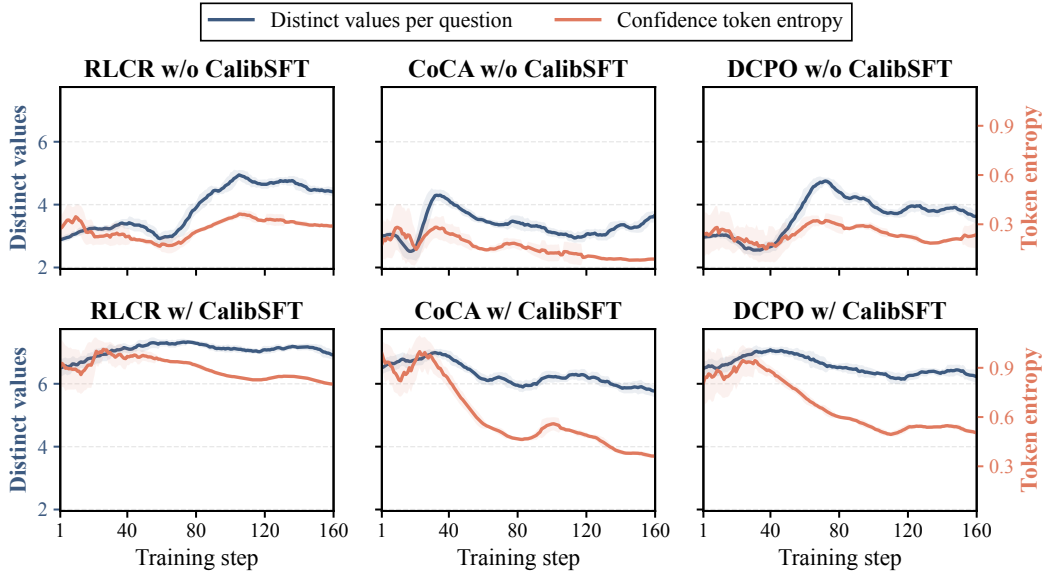


Figure 7: Distinct confidence values per question (left axis) and confidence token entropy (right axis) during training of RLCR, CoCA, and DCPO without (top) and with (bottom) CalibSFT initialization. Each axis uses the same range in all panels. Curves and statistics follow Figure 6.

Table 8: Stability of RLCR on Qwen3-8B across repeated training runs. Each entry is the mean over 2 runs with the standard deviation as a subscript, using the split averages and the evaluation protocol of Table 1. The two rows repeat different stages: RLCR varies the RL seed (44, 45), while + CalibSFT varies the CalibSFT seed (43, 45) and keeps the RL seed fixed, isolating the stage this paper introduces.

Method	In-distribution				Out-of-distribution			
	Pass@1 $\uparrow$	AUROC $\uparrow$	Brier $\downarrow$	ECE $\downarrow$	Pass@1 $\uparrow$	AUROC $\uparrow$	Brier $\downarrow$	ECE $\downarrow$
RLCR	65.80 $\pm$ 0.66	82.08 $\pm$ 0.38	14.64 $\pm$ 0.29	14.26 $\pm$ 0.29	43.53 $\pm$ 0.61	69.05 $\pm$ 0.47	31.67 $\pm$ 0.75	31.99 $\pm$ 1.15
+ CalibSFT	66.55 $\pm$ 0.92	86.19 $\pm$ 0.19	13.38 $\pm$ 0.87	10.36 $\pm$ 2.51	45.69 $\pm$ 0.02	71.77 $\pm$ 0.65	22.70 $\pm$ 0.77	17.75 $\pm$ 0.64

65.90–67.20 with CalibSFT, consistent with CalibSFT improving calibration without trading away accuracy.

D.4 ABLATION STUDIES

Confidence target construction. The construction of confidence targets in CalibSFT is inspired by proper scoring rules, which serve as a foundation for calibration. To verify the effectiveness of this target design, we compare CalibSFT with targets based on group-level correctness (CSFT) (Jang et al., 2025), token probabilities (UaIT) (Liu et al., 2024a), semantic consistency (SaySelf) (Xu et al., 2024), and rubric-based LLM judgments (LoVeC) (Zhang et al., 2026a), using a shared SFT training set and the same optimization objective.

For a fair comparison, we construct the training data from a shared response pool sampled uniformly across empirical success rates so that the comparison covers both easier and harder questions. For each question in DeepScaleR-Train, we use 10 responses generated by Qwen3-8B, with $z_i \in \{0, 1\}$ denoting correctness and $q_x = \frac{1}{10} \sum_{i=1}^{10} z_i$ the empirical success rate. We select 300 questions at each $q_x \in \{0.1, 0.2, \dots, 0.9\}$ and retain one correct and one incorrect response per question, yielding 5,400 responses from 2,700 questions with balanced correctness at each success rate. SaySelf uses only its correct responses following its SFT setup. On this shared subset, we construct the five confidence targets as follows:

- **CSFT.** Both responses receive the group success rate, $c_i^* = q_x$.

Table 9: Training-target distribution (%) over ten equal-width bins in the confidence target ablation.

Target	[0.0, 0.1]	(0.1, 0.2]	(0.2, 0.3]	(0.3, 0.4]	(0.4, 0.5]	(0.5, 0.6]	(0.6, 0.7]	(0.7, 0.8]	(0.8, 0.9]	(0.9, 1.0]
CSFT	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	0.00
UaIT	0.00	0.00	0.00	0.00	0.00	0.00	6.48	56.06	35.15	2.31
SaySelf	22.07	14.41	11.93	10.22	9.04	8.26	8.89	6.63	5.56	3.00
LoVeC	27.33	3.78	8.17	0.20	1.37	0.63	0.22	1.54	15.80	40.96
CalibSFT	11.11	11.11	11.11	11.11	5.56	11.11	11.11	11.11	11.11	5.56

Table 10: AUROC and Brier score on five confidence target construction methods on ID mathematical benchmarks. CalibSFT achieves the better AUROC and Brier score in both phases.

Target	Pre-SFT targets		Post-SFT predictions	
	AUROC $\uparrow$	Brier $\downarrow$	AUROC $\uparrow$	Brier $\downarrow$
CSFT	50.00	31.67	73.46	19.15
UaIT	52.05	33.17	74.72	24.24
SaySelf	—	41.33	57.69	39.38
LoVeC	75.40	24.71	71.11	32.32
CalibSFT	100.00	7.92	79.93	17.30

Table 11: Average calibration performance on eight ID and eight OOD benchmarks across mixing weights. Equal weighting ($\lambda = 0.5$) gives the strongest ID performance.

λ	ID		OOD	
	AUROC $\uparrow$	Brier $\downarrow$	AUROC $\uparrow$	Brier $\downarrow$
0	68.10	32.49	59.20	38.92
0.25	78.02	18.83	66.88	29.94
0.5	79.93	17.30	67.81	30.82
0.75	75.93	20.00	65.82	28.37
1	73.29	19.90	57.22	35.72

- **UaIT.** $c_i^* = \exp(-\text{NLL}_i)$, where NLL_i is the mean negative log-likelihood of the sampled response tokens under the generating model.
- **SaySelf.** We embed the ten responses to each question with Qwen3-Embedding-8B and cluster them at a cosine similarity threshold of 0.98. Each retained correct response is assigned the frequency of its cluster, $c_i^* = |\mathcal{M}_i|/10$, where $\mathcal{M}_i$ is the cluster containing response i .
- **LoVeC.** An LLM judge scores each response on a 0–10 rubric (10: correct with sound reasoning; 8–9: correct with minor gaps; 5–7: partially correct or ambiguous; 2–4: meaningful but incorrect; 0–1: invalid or clearly wrong), normalized as $c_i^* = s_i/10$.
- **CalibSFT.** Following our main design, the target combines group success rate with response correctness, $c_i^* = 0.5q_x + 0.5z_i$.

All variants initialize from Qwen3-8B and use the same supervision objective: reasoning and answer tokens are supervised on correct responses, while confidence tokens are supervised on all responses. We train for 80 steps (40 for SaySelf) with a batch size of 256 and a learning rate of 2×10^{-6} . Table 9 reports the proportion of training targets in each of ten equal-width confidence intervals across the five methods.

Table 10 reports the AUROC and Brier score of the constructed targets before SFT, together with the resulting models’ average performance on in-distribution mathematical benchmarks after SFT. CalibSFT outperforms all four alternatives on both fronts—achieving a lower Brier score and higher AUROC at construction, which transfers to better downstream performance. These results confirm that grounding target construction in proper scoring rules jointly improves calibration and discrimination.

Target mixing weight. We construct confidence targets as $c_i^* = \lambda q_x + (1 - \lambda)z_i$, where q_x is the group success rate and z_i is the correctness of an individual response. To evaluate the effect of the mixing weight, we compare $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$ using the same subset as above and keeping the SFT settings fixed. At $\lambda = 0$, targets are restricted to 0 and 1; this setting yields the highest Brier score in Table 11. At $\lambda = 1$, correct and incorrect responses to the same question receive identical targets, providing no within-question discrimination signal; post-SFT AUROC is also lower than at $\lambda = 0.5$. Equal weighting achieves the highest AUROC and lowest Brier score on ID benchmarks among the tested settings, so we use $\lambda = 0.5$ as the default.

Table 12: Ablation of the CalibSFT supervision design on eight ID benchmarks. The variants differ in the supervised responses for each segment and in loss balancing. CalibSFT achieves the best calibration while keeping Pass@1 close to the best.

Variant	Reasoning/answer		Confidence		Segment balanced	ID performance			
	Correct	Incorrect	Correct	Incorrect		Pass@1 ↑	AUROC ↑	Brier ↓	ECE ↓
Confidence-only	–	–	✓	✓	–	43.56	74.62	20.97	17.43
Full-CE	✓	✓	✓	✓	–	58.49	61.66	29.29	27.65
Full-Balanced	✓	✓	✓	✓	✓	58.46	76.67	19.04	16.87
Correct-only	✓	–	✓	–	✓	59.89	77.58	18.95	16.98
CalibSFT	✓	–	✓	✓	✓	58.51	79.93	17.30	15.95

Correctness-conditional supervision. To verify the effectiveness of CalibSFT design, we ablate three choices in the SFT loss: restricting reasoning and answer supervision to correct responses, supervising confidence on all responses, and balancing the segment losses. Table 12 reports the results. Confidence-only supervision leaves Pass@1 far behind the other variants, showing that reasoning and answer supervision is necessary for accuracy. Without segment balancing, calibration is markedly worse: Full-CE trails Full-Balanced on both AUROC and ECE. Restricting reasoning and answer supervision to correct responses slightly raises accuracy, consistent with incorrect reasoning pulling the model toward incorrect solutions, but restricting confidence supervision to correct responses as well narrows its coverage and weakens calibration. Supervising confidence on all responses while keeping reasoning and answer supervision correct-only recovers this coverage: CalibSFT achieves the best AUROC, Brier score, and ECE among all variants, with Pass@1 close to the best. This combination best balances accuracy, calibration, and confidence diversity, giving a well-calibrated starting point for downstream confidence-aware RL.

D.5 RESULTS UNDER ADDITIONAL METRICS

The main results report Pass@1, AUROC, Brier score, and ECE. We repeat the comparison under two further families of metrics, one for calibration and failure prediction and one for confidence diversity. Both are computed from the responses already sampled for Table 1, so no model is re-evaluated.

Calibration and failure prediction.

- **Positive Calibration Error (PCE)** (Ma et al., 2026) restricts ECE to the bins whose mean confidence exceeds their accuracy, keeping the denominator at the full number of responses, so it isolates the overconfident half of the calibration error.
- **Capability Brier score** (Yang et al., 2026a) scores a question’s mean confidence against its rollout success rate rather than against the correctness of a single response, measuring how well confidence tracks the model’s expected accuracy on that question.
- **E-AURC** (Geifman et al., 2019) subtracts from the area under the risk–coverage curve the area a perfect ranking would attain at the same accuracy. We report it rather than AURC because AURC’s floor is set by accuracy, which Table 1 already reports, and because ranking metrics are otherwise hard to compare across models of differing accuracy (Ding et al., 2020).
- **RCE** (Huang et al., 2024) measures the deviation from a monotone correspondence between confidence and expected correctness, estimated over equal-mass bins, and is designed to be insensitive to both the output range of the confidence measure and the accuracy of the underlying model.

Two caveats apply to the capability Brier score. It estimates each question’s success rate from the four sampled responses, or 32 on AIME, whereas the metric was introduced with 100 samples, so it is a coarse estimate of capability here. It also shares its reference signal with the CalibSFT target $c_i^* = \lambda q_x + (1 - \lambda) z_i$, which is built from the training-set success rate; the two are computed on

Table 13: Calibration and failure prediction under additional metrics, on the in-distribution and out-of-distribution splits and under the protocol of Table 1. PCE is the part of ECE contributed by the overconfident bins, Cap. Brier scores each question’s mean confidence against its rollout success rate, E-AURC is the excess of the risk–coverage area over a perfect ranking, and RCE measures the deviation from a monotone correspondence between confidence and expected correctness. All values are percentages.

Method	In-distribution				Out-of-distribution			
	PCE ↓	Cap. Brier ↓	E-AURC ↓	RCE ↓	PCE ↓	Cap. Brier ↓	E-AURC ↓	RCE ↓
RLCR	9.71	10.63	7.75	13.50	30.99	27.53	19.74	11.45
+ CalibSFT	7.30	8.66	5.61	5.30	14.77	18.72	17.14	4.93
ReDoubt	6.71	11.98	8.02	15.23	22.51	21.71	18.34	14.77
+ CalibSFT	5.54	9.33	4.77	7.69	15.58	18.80	15.85	4.70
CoCA	10.96	12.82	8.88	19.49	22.64	22.84	21.52	15.13
+ CalibSFT	11.40	12.24	5.50	8.58	13.39	19.41	17.82	5.79
DCPO	10.04	10.95	10.10	18.79	31.97	28.95	19.62	15.40
+ CalibSFT	8.09	9.00	5.92	9.21	12.95	17.54	16.23	6.18
C3RL	28.01	20.75	16.11	33.15	47.96	41.92	25.22	27.57
+ CalibSFT	10.72	10.82	8.76	17.24	8.80	18.46	21.27	19.08

disjoint questions, but the metric is by construction aligned with what CalibSFT supervises, so we read it alongside the other three rather than on its own.

Table 13 shows the same pattern as the main results. Across the five RL objectives, the two splits, and the four metrics, CalibSFT initialization wins 39 of the 40 comparisons. The one exception is the in-distribution PCE of CoCA, which rises from 10.96 to 11.40. The largest and most consistent gap is in RCE: out of distribution it falls from 11.45 to 4.93 for RLCR, from 14.77 to 4.70 for ReDoubt, from 15.13 to 5.79 for CoCA, and from 15.40 to 6.18 for DCPO, roughly a third of its value in each case. Since RCE is constructed to be independent of the accuracy of the underlying model, this gap cannot be attributed to the accuracy differences between the two initializations. PCE moves in the same direction and by a wide margin out of distribution, falling from 31.97 to 12.95 for DCPO and from 47.96 to 8.80 for C3RL, which shows that the calibration gains of CalibSFT come from removing overconfidence rather than from trading it against underconfidence.

Confidence diversity.

- **Distinct** counts the confidence values a model produces over the responses of a split, measuring the size of the support regardless of how often each value is used.
- **Top-1** and **Top-3** are the share of responses taking the most frequent one and three values, the same quantity reported for the confidence priors in Figure 1.
- **Effective** is the exponential of the entropy of the value distribution, that is, the number of values that would carry the same mass if they were used uniformly. It complements Distinct, which a value used once inflates as much as a value used by half the responses.

We do not report the mass at the ends of the scale or the fraction of occupied confidence bins. The former is near zero for most runs, since the concentrated priors of Figure 1 sit below the ceiling rather than at it, and the latter reduces to Distinct once the bins are fine enough to separate values on the two-decimal grid the models emit.

Table 14 shows that CalibSFT broadens confidence support for every objective and on both splits, with no exceptions among the 40 comparisons. The clearest case is RLCR out of distribution, where the number of distinct values rises from 18.62 to 94.50 while the three most frequent values fall from 66.22% of responses to 9.90%, and the effective number of values rises from 7.03 to 72.83. Distinct and Effective move together throughout, so the broader support is not an artifact of rarely used values: the added values carry mass. C3RL is the one objective whose support stays narrow, at 10.00 distinct values, yet even there the most frequent value drops from 71.04% to 36.37% of responses, which raises its effective support from 2.79 to 6.46. Read together with Table 13, the objectives that gain the

Table 14: Confidence diversity on the in-distribution and out-of-distribution splits, under the protocol of Table 1. Distinct counts the confidence values a model produces, Top-1 and Top-3 are the share of responses taking its most frequent one and three values, and Effective is the exponential of the entropy of the value distribution, that is, the number of values that would carry the same mass if used uniformly. Distinct and Effective are counts; Top-1 and Top-3 are percentages.

Method	In-distribution				Out-of-distribution			
	Distinct $\uparrow$	Top-1 $\downarrow$	Top-3 $\downarrow$	Effective $\uparrow$	Distinct $\uparrow$	Top-1 $\downarrow$	Top-3 $\downarrow$	Effective $\uparrow$
RLCR	15.75	33.30	67.74	6.90	18.62	33.18	66.22	7.03
+ CalibSFT	86.88	12.59	20.57	49.42	94.50	3.73	9.90	72.83
ReDoubt	7.62	43.17	92.47	3.50	10.88	44.84	86.31	4.20
+ CalibSFT	58.00	9.43	23.64	32.03	59.50	7.83	20.70	31.62
CoCA	12.75	55.50	77.38	4.85	13.62	46.41	67.06	6.06
+ CalibSFT	47.62	13.62	30.12	26.22	52.12	10.71	26.88	24.69
DCPO	15.75	59.32	80.94	4.40	18.62	43.97	73.64	6.06
+ CalibSFT	72.75	21.54	41.93	27.14	84.25	5.78	15.07	53.93
C3RL	8.50	83.23	96.46	1.97	9.38	71.04	91.77	2.79
+ CalibSFT	10.00	46.09	70.97	5.68	10.00	36.37	65.54	6.46

most support are also those whose RCE improves the most, consistent with the exploration argument of Section 3.

D.6 INTEGRATION WITH OTHER CONFIDENCE ESTIMATORS

A model can be scored with several confidence estimators, of which the verbalized value is only one. To test whether the calibration that CalibSFT provides also reaches the others, we evaluate two widely used estimators on the responses sampled for the main evaluation, so that the answers, their correctness, and the sampling protocol match Table 1.

- **Token probability.** We read the next-token distribution at the position of the confidence value. Restricted to the ten digit tokens, the distribution after `<confidence>` gives the integer digit and the distribution after `<confidence>0.` gives the first decimal, which we combine into $c = P(1) + P(0) \sum_{d=0}^9 p(d) (d/10 + 0.05)$, replacing the unobserved second decimal by the midpoint of its interval. Both distributions are taken at temperature one; using the sampling temperature 0.6 instead changes the averages by at most 0.5 points. Responses whose confidence was filled in after the generation limit, at most 0.42% of the responses of a benchmark, keep their verbalized value.
- **Self-consistency.** We group the four responses to a question, 32 on AIME, by answer equivalence and keep the largest group, with ties broken by sampling order. The answer of that group determines correctness and the mean verbalized confidence of its responses becomes the confidence, so this estimator scores one prediction per question and its accuracy column reports majority-vote accuracy.

Table 15 reports both estimators next to verbalized confidence. CalibSFT initialization improves every metric under every estimator on both splits. Under token probability, AUROC rises from 83.60 to 87.66 in-distribution and Brier score falls from 31.22 to 22.80 out-of-distribution; under self-consistency, ECE falls from 17.62 to 9.67 and from 30.53 to 18.19. The estimators themselves differ in what they add. Token probability raises AUROC by about 1.3 points in-distribution over the verbalized value it is read from, for both initializations. Self-consistency raises majority-vote accuracy and lowers Brier score, but its ECE stays above the verbalized value in-distribution, 17.62 against 14.46 without CalibSFT and 9.67 against 8.59 with it.

D.7 CONFIDENCE VISUALIZATION

Figures 8 and 9 show reliability diagrams of RLCR with and without CalibSFT initialization on three in-distribution and three out-of-distribution benchmarks, computed from the responses of the main evaluation over ten equal-width confidence intervals.

Table 15: Confidence estimators for RLCR with and without CalibSFT initialization, averaged over the eight in-distribution and eight out-of-distribution benchmarks. CalibSFT initialization improves calibration and discrimination under all three estimators. Verbalized and token-probability confidence score the same responses and therefore share Pass@1, while self-consistency aggregates those samples by majority vote, so its accuracy column reports majority-vote accuracy.

Estimator	Method	In-distribution				Out-of-distribution			
		Pass@1 $\uparrow$	AUROC $\uparrow$	Brier $\downarrow$	ECE $\downarrow$	Pass@1 $\uparrow$	AUROC $\uparrow$	Brier $\downarrow$	ECE $\downarrow$
Verbalized	RLCR	65.33	82.36	14.43	14.46	43.96	69.38	31.14	31.17
	+ CalibSFT	67.20	86.32	12.77	8.59	45.71	71.31	23.24	18.20
Token probability	RLCR	65.33	83.60	14.35	15.03	43.96	69.70	31.22	30.54
	+ CalibSFT	67.20	87.66	12.75	10.13	45.71	71.51	22.80	19.29
Self-consistency	RLCR	70.86	85.36	13.09	17.62	44.83	70.83	30.61	30.53
	+ CalibSFT	72.31	87.64	11.35	9.67	47.24	72.51	22.77	18.19

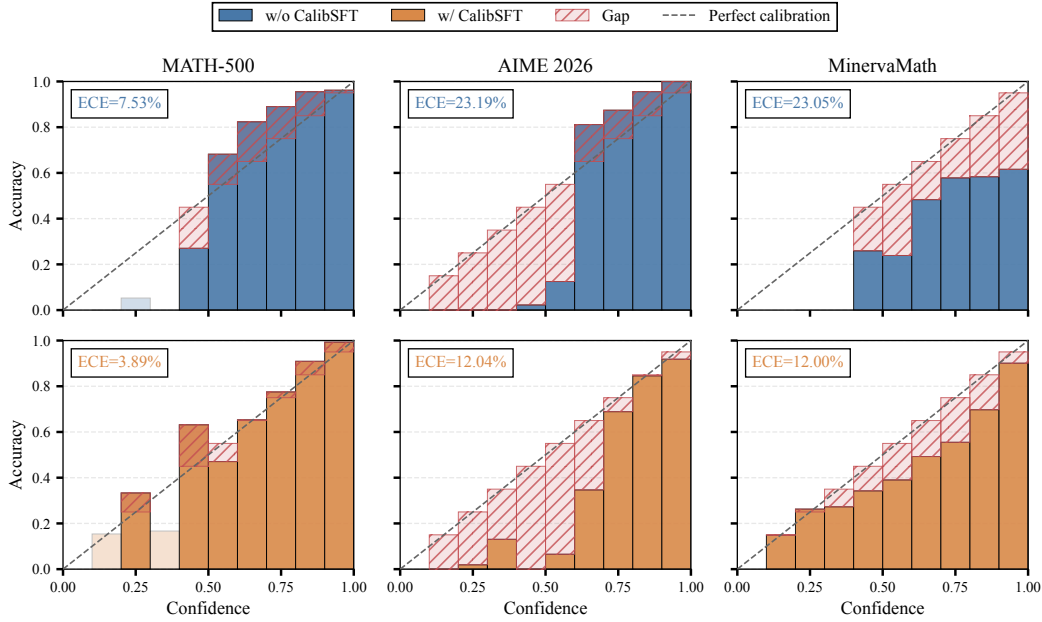


Figure 8: Reliability diagrams of RLCR with and without CalibSFT initialization on in-distribution benchmarks. Bars show the accuracy in each of ten equal-width confidence bins, and hatched regions show the gap to the bin midpoint. Bins containing less than 1% of responses are faded and shown without gaps.

CalibSFT reduces miscalibration in both directions. Where RLCR is overconfident, its mean confidence exceeds its accuracy by 23.0 points on MinervaMath, 52.4 on NQOpen, and 29.1 on LiveBenchReasoning, and the bars move toward the diagonal once RL starts from CalibSFT: ECE falls from 23.05 to 12.00 on MinervaMath and from 52.43 to 30.89 on NQOpen. Where RLCR is underconfident, on MATH-500, its accuracy exceeds its confidence for 97% of the responses, and CalibSFT lowers ECE from 7.53 to 3.89. The correction also spreads the responses over more intervals: on TriviaQA the share in the top interval falls from 39.0% to 16.4% while ECE falls from 23.30 to 10.71, and the number of intervals holding at least 1% of the responses rises from six to nine on MinervaMath and NQOpen and from seven to nine on TriviaQA. What remains differs by benchmark, as NQOpen stays overconfident in every occupied interval, while LiveBenchReasoning turns underconfident for 95% of its responses.

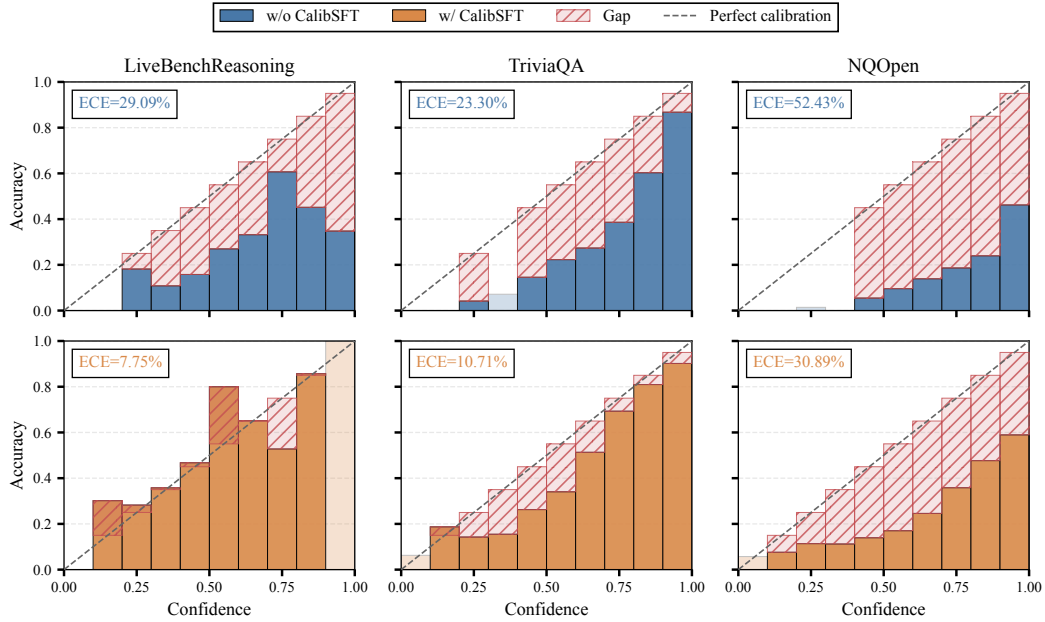


Figure 9: Reliability diagrams of RLCR with and without CalibSFT initialization on out-of-distribution benchmarks. Bars, gaps, and faded bins follow Figure 8.

Table 16: Training cost of the CalibSFT stage and of confidence-aware RL on one node with eight NVIDIA B20Z GPUs. Runs of 180 (CalibSFT) and 160 (RL) optimization steps are scaled linearly to a common budget of 200 steps.

Stage	Minutes per step	Wall-clock (h)	GPU-hours
CalibSFT	0.33	1.1	8.8
RLCR	4.77	15.9	127.3
RLCR + CalibSFT	4.58	15.3	122.2

D.8 ANNOTATION AND COMPUTE COST

CalibSFT requires no human annotation. Its targets combine the group success rate q_x with the correctness z_i of an individual response, and both come from the verifier that also produces the reward during RL, applied to responses the model samples itself. The responses are drawn once, before training, and the resulting targets initialize every confidence-aware RL method we study, whereas RL draws new responses at every optimization step.

Table 16 compares the two training stages at a common budget of 200 optimization steps, each run on one node with eight NVIDIA B20Z GPUs. A step of CalibSFT takes 0.33 minutes against 4.77 minutes for a step of RLCR, which samples new responses before every update, so the supervised stage costs 6.9% of the GPU-hours the RL stage does. Wall-clock time includes model loading, periodic validation, and checkpoint writing.

E DOWNSTREAM APPLICATIONS

E.1 SELECTIVE RISK CONTROL

A model that reports calibrated confidence can decide when to answer and when to abstain, but a deployment needs that decision to come with a guarantee rather than with a threshold chosen by hand. Learn-then-Test (Angelopoulos et al., 2025) turns the choice into a multiple-testing problem: for

each candidate threshold, the null hypothesis states that the selective risk, the error rate among the responses the model answers, exceeds a budget $r_{\max}$, and any threshold whose hypothesis is rejected carries a finite-sample guarantee. Following Shen (2026), we test the hypotheses with Clopper–Pearson upper limits (Clopper & Pearson, 1934) and correct for the 101 candidate thresholds on the grid $\{0, 0.01, \dots, 1\}$ with a union bound at $\delta = 0.05$, and we keep the smallest threshold that passes, which answers as many questions as the guarantee allows. Conformal risk control (Angelopoulos et al., 2024) offers a simpler construction, but it bounds the expectation of a monotone loss, whereas the selective risk is a ratio whose denominator depends on the threshold and the guarantee we want holds with probability $1 - \delta$ for the single threshold a deployment fixes. We split the questions of a split in half. The calibration half keeps one response per question, so the counted responses are independent draws, and the test half contributes all of its responses. The same machinery supports other decisions once a confidence score is available, such as routing a question to a set of models under a misrouting-risk constraint (Hao et al., 2026) or accepting model labels under a false-discovery-rate constraint (Huang et al., 2025). Validity alone does not make a threshold useful: a confidence that takes few distinct values leaves few candidates that can pass, so the certificate becomes conservative even when the ranking is informative.

Table 17 reports the coverage that this procedure certifies. At a 10% risk budget in-distribution, CalibSFT initialization raises certified coverage for every method, from 31.4 to 62.6 for RLCR and from 28.8 to 66.1 for ReDoubt, and it certifies CoCA and C3RL at 52.3 and 49.2 where no threshold passes without it. Out-of-distribution the separation is sharper: at a 20% budget none of the five methods admits any certified threshold, while CalibSFT initialization certifies all five between 16.8 and 22.8. The remaining gap to a perfect ranking of the same responses, 82.0 in-distribution and 54.5 out-of-distribution, shows how much of the loss comes from the task rather than from the procedure; at a 10% budget out-of-distribution, where a perfect ranking would certify only 48.4, no configuration passes. At a 10% budget in-distribution and at a 20% budget out-of-distribution, the realized error rate on held-out questions stays below the budget in every configuration, reaching at most 9.4 and 18.2. At a 20% budget in-distribution four configurations exceed it on a minority of the calibration splits, by at most 3.5 points, which is what a guarantee that holds with probability $1 - \delta$ allows. Figure 10 sweeps the budget and shows where the two initializations differ. Without CalibSFT, certified coverage climbs in steps and jumps almost vertically for CoCA and C3RL, because the few confidence values the model emits leave few usable thresholds; the same sparsity makes the baselines vary across calibration splits, at a 20% budget in-distribution by 35.3 points for C3RL and 12.3 for ReDoubt against 0.5 and 1.2 with CalibSFT. With CalibSFT the curve rises smoothly, so a threshold exists for any budget a deployment may ask for.

E.2 CONFIDENCE-GUIDED MODEL ROUTING

A calibrated confidence also decides which questions a small model should answer itself and which it should pass to a stronger one, and a deployment needs that decision to come with a bound on what the routing costs in accuracy. Each local policy of Table 1 answers every question first, which is where its confidence comes from, and a question whose confidence falls below a threshold is answered again by Qwen3.7-Flash, a hosted reasoning model; the cost a threshold saves is therefore the hosted model’s, as the local answer is always produced. Following the performance guarantees established for efficient reasoning (Yu et al., 2026a; Zeng et al., 2025) and for risk-aware routing (Hao et al., 2026), the threshold is selected on a calibration split so that the accuracy lost against deferring every question stays below a tolerance ε with probability $1 - \delta$. That loss is signed, since the local model is sometimes right where Qwen3.7-Flash is wrong, and it takes values in $[-1, 1]$: a Hoeffding bound on its mean leaves a slack of 7.9 points on our 2,414 calibration questions, so no tolerance near 1% would be testable. We therefore bound only the harmful event, an accepted question the local model answers incorrectly while Qwen3.7-Flash answers it correctly. Its rate upper bounds the loss, ignoring the questions where the local model is the one that is right, and it is a Bernoulli mean, so we test it with a Clopper–Pearson limit (Clopper & Pearson, 1934) and a union bound over the 102 candidate thresholds at $\delta = 0.05$, as in Appendix E.1; with no harmful event observed the limit reaches 0.32%, which is what makes a 1% tolerance attainable at all. The pool is the eight mathematics benchmarks of the in-distribution split that Qwen3.7-Flash answered, 4,828 questions, and each question contributes one response, so the counted events are independent draws.

Table 18 reports what the certified threshold costs. At a 1% tolerance CalibSFT initialization lowers the deferral rate for every method, by 12.3 points on average, and at a 2% tolerance for three of the

Table 17: Certified coverage (%) of confidence-aware RL with and without CalibSFT initialization. A threshold is selected on half of the questions so that the error rate among answered responses stays below the risk budget with probability 0.95, and the coverage it reaches on the held-out questions is averaged over ten splits. The last row is the coverage a perfect ranking of the same responses would certify, averaged over the ten models; it varies by at most 2.9 points between them.

Method	In-distribution		Out-of-distribution
	$r_{\max} = 10\%$	$r_{\max} = 20\%$	$r_{\max} = 20\%$
RLCR	31.4 $\pm$ 0.6	86.4 $\pm$ 1.1	0.0 $\pm$ 0.0
+ CalibSFT	62.6$\pm$2.5	88.6$\pm$1.3	20.3$\pm$0.8
ReDoubt	28.8 $\pm$ 0.5	70.2 $\pm$ 12.3	0.0 $\pm$ 0.0
+ CalibSFT	66.1$\pm$0.9	88.6$\pm$1.2	18.0$\pm$2.5
CoCA	0.0 $\pm$ 0.0	82.3 $\pm$ 1.4	0.0 $\pm$ 0.0
+ CalibSFT	52.3$\pm$10.5	87.2$\pm$2.5	20.5$\pm$0.2
DCPO	1.6 $\pm$ 4.7	85.5 $\pm$ 2.2	0.0 $\pm$ 0.0
+ CalibSFT	33.3$\pm$21.4	87.3$\pm$1.4	22.8$\pm$1.5
C3RL	0.0 $\pm$ 0.0	17.7 $\pm$ 35.3	0.0 $\pm$ 0.0
+ CalibSFT	49.2$\pm$16.4	84.8$\pm$0.5	16.8$\pm$0.2
Perfect ranking	82.0	92.3	54.5

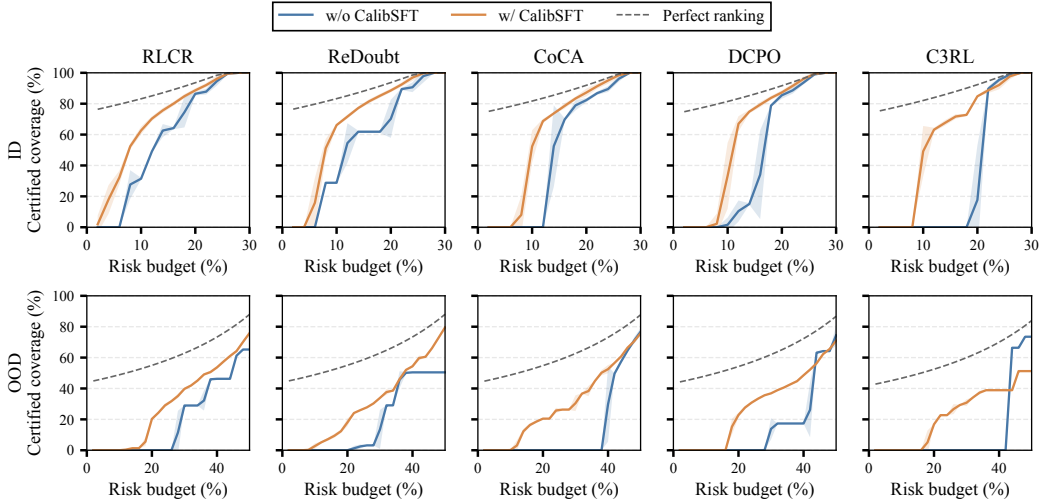


Figure 10: Certified coverage across risk budgets for confidence-aware RL with and without CalibSFT initialization, on the in-distribution (top) and out-of-distribution (bottom) benchmarks. Curves average ten calibration splits and shaded bands show their standard deviation. The dashed line is the coverage a perfect ranking of the same responses would certify.

four, by 25.1 points, while the accuracy stays within 0.8 points of the 82.58% that deferring every question reaches. The baselines are not merely more expensive but close to degenerate: CoCA defers 98.7% of the questions at both tolerances and DCPO 82.9%, so the routing avoids 0.6% and 5.6% of the Qwen3.7-Flash tokens, against 33.1% and 43.5% once RL starts from CalibSFT at the 2% tolerance. The sparsity that leaves few usable thresholds in Appendix E.1 leaves few usable thresholds here as well. Counting questions also understates what routing costs, because the questions a model is unsure about are the ones that take the longest to answer: RLCR defers 45.9% of the questions at the 2% tolerance yet avoids only 23.3% of the Qwen3.7-Flash tokens, and with CalibSFT initialization it defers 39.2% and avoids 36.1%.

Table 18: Certified deferral to a stronger model for confidence-aware RL with and without CalibSFT initialization. The local model answers a question when its confidence reaches a threshold and defers it to the hosted model otherwise. The threshold is selected on half of the questions so that the accuracy lost against deferring every question stays below the tolerance with probability 0.95, and what it reaches on the held-out questions is averaged over ten splits. The certificate constrains accuracy, so the accuracy column records that the guarantee was met rather than a quantity being compared; what the threshold trades is cost, reported as the share of questions deferred and the share of the hosted model’s tokens it avoids against deferring every question. Deferring every question answers 82.58% correctly and deferring none answers between 76.3% and 78.0% correctly.

Method	Tolerance $\varepsilon = 1\%$			Tolerance $\varepsilon = 2\%$		
	Deferred (%) $\downarrow$	Tokens saved (%) $\uparrow$	Accuracy	Deferred (%) $\downarrow$	Tokens saved (%) $\uparrow$	Accuracy
RLCR	65.0 $\pm$ 1.1	10.1	82.97	45.9 $\pm$ 1.0	23.3	83.33
+ CalibSFT	60.3 $\pm$ 3.6	17.9	82.66	39.2 $\pm$ 1.0	36.1	82.47
ReDoubt	67.4 $\pm$ 0.8	10.1	82.99	43.1 $\pm$ 1.0	27.1	83.24
+ CalibSFT	55.3 $\pm$ 2.7	23.8	82.81	36.3 $\pm$ 3.7	42.1	82.60
CoCA	98.7 $\pm$ 0.1	0.6	82.61	98.7 $\pm$ 0.1	0.6	82.61
+ CalibSFT	71.7 $\pm$ 4.9	17.3	82.51	51.2 $\pm$ 5.5	33.1	82.42
DCPO	82.9 $\pm$ 0.7	5.6	82.64	82.9 $\pm$ 0.7	5.6	82.64
+ CalibSFT	77.4 $\pm$ 0.3	12.2	82.58	33.3 $\pm$ 3.8	43.5	82.80